6 Sequential Circuits

Introduction

In combination circuits, present output depends only upon the present input, but in sequential circuits present output depends upon present input as well as past output (present state). So, the memory element in the sequential circuit stores the past output and provide that stored output to output logic.

Block diagram of sequential circuits:

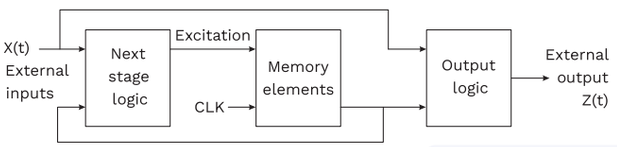


Fig. 6.1 Block Diagram of Sequential Circuits

In the sequential circuit memory element work only when the clock signal (CLK) is applied, it means a clock signal is a must for any sequential circuit.

Comparison between combinational and sequential circuits:

S. No.	Combinational Circuits	Sequential Circuits
1.	In a combinational circuit, the output at any instant of time depends only on the present input variables.	In sequential circuits, the output at any instant of time depends not only on the present input, but also on the present state, i.e., the present state or the past history of the system.
2.	Memory unit is not required.	Memory unit is required to store the past history of the input.
3.	Combinational circuits are faster.	Sequential circuits are slower than combinational circuits.
4.	Combinational circuits are easy to design.	Sequential circuits are comparatively harder to design.

Table 6.1

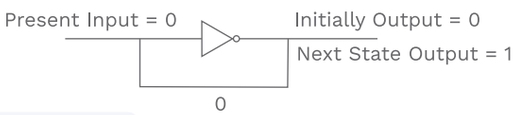


Fig. 6.2 NOT Gate Circuit

In the above NOT gate circuit, let us assume the initially output is logic '0'. So this output comes to an input terminal of the circuit, hence the present input will become logic '0' and the next state output of the circuit will be logic '1'.

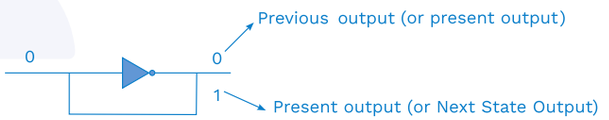


Fig. 6.3 NOT Gate Circuit Outputs

Q_n = Q→ Past output (Present output)

Q_{n+1} = Q^+→ Present output (Next output)

- The logic circuit whose output at any instant of time depends on the present inputs as well as in the past outputs are called sequential circuits. In the sequential circuit, the output signals are feedback to the input side.

Latch:

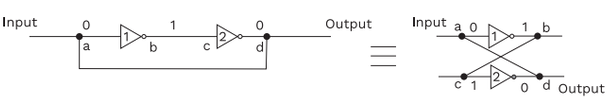


Fig. 6.4 Latch

Input = Output = logic '0'

In the given circuit, provided input will be stored inside the circuit. So we can remove the input and data will be stored forever. This circuit can store only 1 bit of data and works as a memory element. So this circuit is called 'Latch'.

NAND Latch:

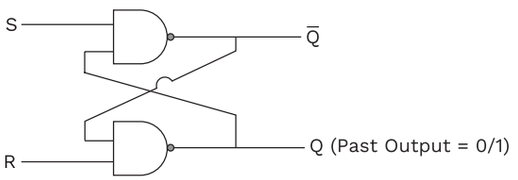


Fig. 6.5 NAND Latch

- In the above circuit, past output (Q) may be logic '0' or logic '1'. Here present output depends upon the present input as well as past output. We can use the table to define the relation between inputs and outputs.

Previous Years' Questions

Choose the correct statements relating to the circuit of figure.

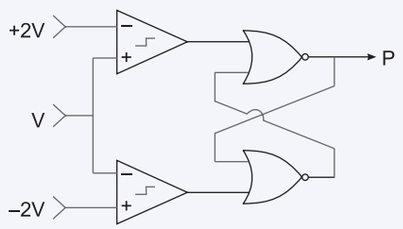


Fig. 6.6

(A) For V1 = –2V, P = 0
(B) For V1 = +3V, P = 0
(C) V1 = 0V, P = 0 always
(D) For V1 = 0V, P can be either 0 or 1

Solution: (B)

Present Input		Past Output	Present Output	
S	R	Q	Q^+	$\overline{Q^+}$
0	0	0	1	1
0	0	1	1	1
0	1	0	0	1
0	1	1	0	1
1	0	0	1	0
1	0	1	1	0
1	1	0	0	1
1	1	1	1	0

Table 6.2 Truth Table of NAND Latch

- If input S = 0 and R = 0, then present output (Q+) becomes logic '1' for the past output Q = 0 and Q = 1 and NAND latch goes into the invalid state.
- If input S = 0 and R = 1, then the present output (Q+) reset (logic '0') for the past output Q = 0 and Q = 1.
- If input S=1 and R=0, then the present output (Q+) set (logic '1') for the past output Q=0 and Q=1.
- If maintain a gap S=1 and R=1, then the present output 'Hold' the state of the past output (for Q = 0, Q+ = 0 and for Q = 1, Q+=1).

S	R	Q^+	State
0	0	$Q^+ = \overline{Q^+} = 1$	Invalid
0	1	0	Reset
1	0	1	Set
1	1	Q	Hold

Table 6.3

Previous Years' Questions

An R-S latch is:
(A) Combinational circuit
(B) Synchronous sequential circuit
(C) One bit memory element
(D) One clock delay element

Solution: (C)

NOR Latch

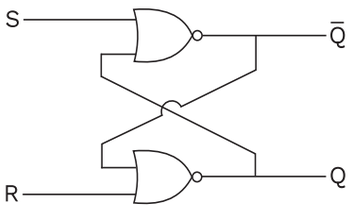


Fig. 6.6 NOR Latch

S	R	Q	Q^+	$\bar{Q}^+$
0	0	0	0	1
0	0	1	1	0
0	1	0	0	1
0	1	1	0	1
1	0	0	1	0
1	0	1	1	0
1	1	0	0	0
1	1	1	0	0

Table 6.4

- If S = 0 and R = 0, then the present output 'Hold' the state of past output (for Q = 0, Q+ = 0 and Q = 1, Q+ = 1).
- If S = 0 and R=1, then the present output (Q+) reset (logic '0') for past output Q = 0 and Q =1.
- If S = 1 and R=0, then the present output (Q+) set (logic '1') for past output Q = 0 and Q =1.
- If S=1 and R = 1, then the present output (Q+) become (logic '0') for past output Q=0 and Q =1 and NOR latch goes into the invalid state.

S	R	Q^+	State
0	0	Q	Hold
0	1	0	Reset
1	0	1	Set
1	1	$Q^+ = \bar{Q}^+ = 0$	Invalid

Table 6.5

Previous Years' Questions

The circuit in the figure has two CMOS NOR gates. This circuit functions as a:

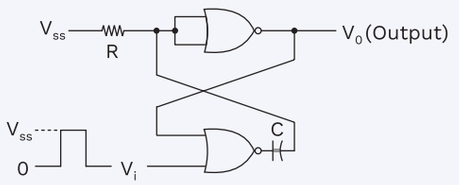


Fig. 6.7

(A) Flip-flop
(B) Schmitt trigger
(C) Monostable multivibrator
(D) Astable multivibrator

Solution: (C)

Example 1:

Assume initially Q = 0 for the given circuit, then find the output for the following sequence, SR: 01→10→11→00→01→11→10.

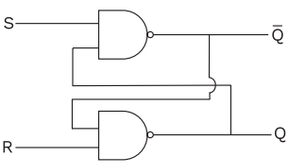


Fig. 6.8 SR Flipflop

Solution:

Initially output Q = 0

Truth table:

S	R	Q^+	
–	–	0	← Initially
0	1	0	
1	0	1	Hold
1	1	1	
0	0	1	
0	1	0	Hold
1	1	0	

Table 6.6

- Output of the combinational circuits depends upon the input combination.
- Output of the sequential circuits depends upon the input sequence.
- If there is an unequal delay in gates, the given operation is also valid.

Example 2:

Assume initially Q=0 for the given circuit (with equal NAND gate delays of 2 n sec), then find the output for the following sequence (SR): 00--->>11.

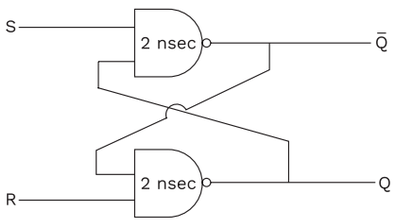


Fig. 6.9

Solution:

Initially Q = 0 and $\bar{Q} = 1$, if we put the input 00 then 11, the output of the circuit depends on whether the NAND gate delays are equal or not.

For Equal Delay (2 N Sec):

Delay	S	R	Q	$\bar{Q}$
Initially	-	-	0	1
2 nsec	0	0	1	1
4 nsec	1	1	0	0
6 nsec	1	1	1	1
8 nsec	1	1	0	0
10 nsec	1	1	1	1
12 nsec	1	1	0	0
:	:	:	:	:
:	:	:	:	:

Table 6.7

In the given circuit, for input S=1 and R=1 output changes their state with time. So this circuit is unstable and this problem is called a **critical race**.

Previous Years' Questions

Refer to the NAND and NOR latches are shown in the figure. The input (P1, P2) for both the latches are first made (0, 1) and then, after a few seconds, made (1, 1). The corresponding stable output (Q1, Q2) are:

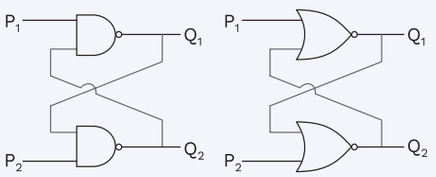


Fig. 6.10

(A) NAND: first (0, 1) then (0, 1) NOR first (1, 0) then (0, 0)
(B) NAND: first (1, 0) then (1, 0) NOR first (1, 0) then (1, 0)
(C) NAND: first (1, 0) then (1, 0) NOOR: first (1, 0) then (0, 0)
(D) NAND: first (1, 0) then (1, 1) NOR: first (0, 1) then (0, 1)

Solution: (C)

Example 3:

Assume initially Q=0 for the given circuit (with unequal NAND gate delays of 2 nsec and 4 nsec), then find the output for the following sequence (SR): 00--->> 11.

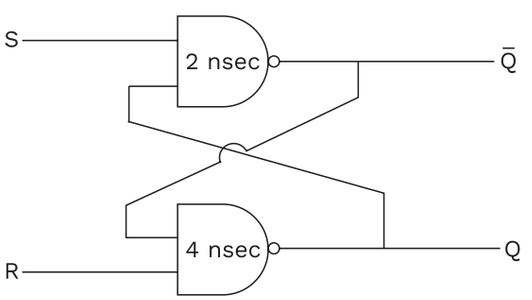


Fig. 6.11

Solution:

Initially, Q=0 and $\overline{Q} = 1$, if we put the input 00 then 11 output of the circuit depends on whether the NAND gate delay is equal or not.

Delay	S	R	Q	$\overline{Q}$
Initially	-	-	0	1
2 nsec	0	0	1	1
4 nsec	0	0	1	1
6 nsec	1	1	0	1

→ Output is stable. So the changing input 00 → 11

Table 6.8

So the given circuit is stable circuit.

1. For SR NAND latch, if the input sequence 00 is followed by 11, then the following case arises,
   a) If the delay of both gates are same, then we don't have any stable output, the output is oscillatory, this condition is known as 'critical race.'
   b) However, if the delay of both gates are not equal then there exists a stable output, but it depends on the individual delay of the gates.

Example 4:

Assume initially Q = 0 for given circuit, then find the output of the following sequence (SR): 01-->10-->00-->11-->01-->00-->10

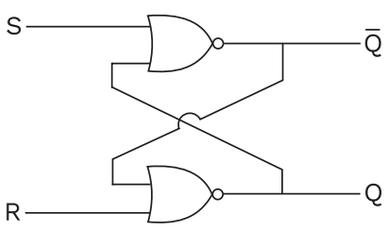


Fig. 6.12

Solution:

Initially output Q=0,

S	R	Q	
–	–	0	
0	1	0	
1	0	1	Hold
0	0	1	
1	1	0	
0	1	0	Hold
0	0	0	
1	0	1	

Table 6.9

The output of sequential circuits depends on the sequence of input.

Similarly, for SR NOR latch, if the input sequence 11 is followed by 00, then the following case arises,

1. If the delay of both gates is same, then we don't have any stable output, the output is oscillatory, this condition is known as critical race.
2. However, if the delay of both gates, are not equal then there exists a stable output, but it depends on the individual delay of the gates.

Flip Flop

- In a latch, the output changes immediately in response to external input, so to an additional control on the latch, we are introducing a signal called 'CLOCK', whose purpose is the same as enable pin of Decoder.
- Flip-flop is the combination of latch and clock.

Latch + Clock = Flip Flop

Previous Years' Questions

The following binary values were applied to the X and Y inputs of the NAND latch shown in the figure in the sequence indicated below:
X = 0, Y = 1; X = 0, Y = 0; X = 1, Y = 1.
The corresponding stable P, Q outputs will be

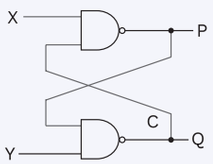


Fig. 6.13

(A) P=1, Q=0, P=1, Q=0 or P=0, Q=1
(B) P=1, Q=0; P=1, Q=0 or P=0, Q=1; P=0, Q=1
(C) P=1, Q=0, P=1, Q=1; P=1, Q=0 or P=0, Q=1
(D) P=1, Q=0; P=1, Q=1; P=1, 0=1

Solution: (C)

SR Flip-Flop

1. SR Flip-Flop using NAND Latch:

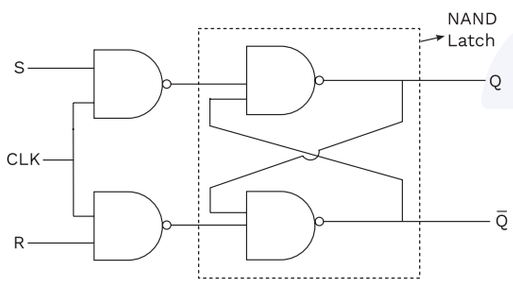


Fig. 6.14 SR Flip flop Using NAND Latch

a) In SR flip-flop, using NAND latch, If CLK is logic '0' than the present output (Q+) of flip-flop 'Hold' the state of the past output (Q), irrespective of whatever input we will provide at S and R terminals.

$Q^+ = Q$

CLK	S	R	Q	Q^+
0	X	X	Q	Q
1	0	0	0	0
1	0	0	1	1
1	0	1	0	0
1	0	1	1	0
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

Table 6.10

Previous Years' Questions

For the circuit shown in the figure, D has a transition from 0 to 1 after CLK changes from 1 to 0. Assume gate delays to be negligible.

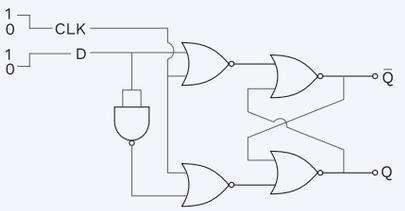


Fig. 6.15

Which of the following statement is true?
(A) Q goes to 1 at the CLK transition and stays at 1
(B) Q goes to 0 at the CLK transition and stays at 0
(C) Q goes to I at the CLK transition and goes to 0 when D goes to 1
(D) Q goes to 0 at the CLK transition and goes to I when D goes to 1.

Solution: (C)

2. SR Flip-Flop Using Nor Latch:

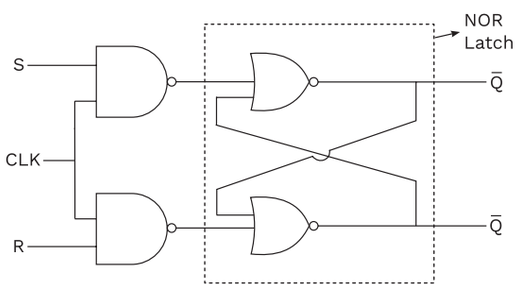


Fig. 6.16 SR Flip-Flop using NOR-Latch

In SR flip-flop using NOR Latch, if CLK is logic '0' then the present output (Q^+) of flip-

flop 'Hold' the state of the past output (Q), irrespective of whatever the input we will provide at S and R terminals.

CLK	S	R	Q	Q^+
0	X	X	Q	Q
1	0	0	0	0
1	0	0	1	1
1	0	1	0	0
1	0	1	1	0
1	1	0	0	1
1	1	0	1	1
1	1	1	0	0
1	1	1	1	0

Table 6.11

a) Latches are universally not unique, hence their truth table is not unique.
b) Flip-flop is universally unique, hence there truth tables are unique.

SR Flip Flop Operation:

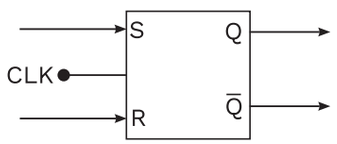


Fig. 6.17 SR Flip-flop Operation

Previous Years' Questions

Consider the D-Latch shown in the figure, which is transparent when its clock input CLK is high and has zero propagation delay. In the figure, the clock signal CLK1 has a 50% duty cycle and CLK2 is a one-fifth period delayed version of CLK1. The duty cycle at the output of the latch in percentage is:

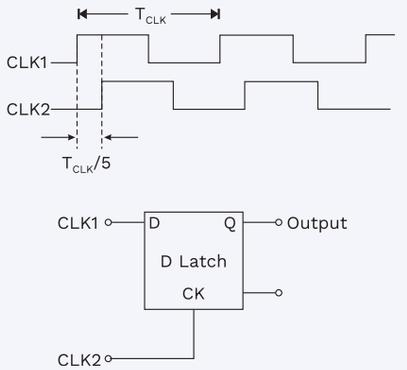


Fig. 6.18

Solution: 30

- If CLK = 0, irrespective of inputs (S or R) present output will hold past the output state ($Q^+ = Q$).
- If CLK = 1 and S = 0, R = 0, then the present output will hold past the output state ($Q^+ = Q$).
- If CLK = 1 and S = 0, R = 1 then the present output (Q^+) will reset (logic '0'), irrespective of the past output (Q).
- If CLK = 1 and S = 1, R = 0 then the present output (Q^+) will set (logic '1'), irrespective of the past output (Q).
- If CLK = 1 and S = 1, R = 1 then the present output (Q^+) goes into the invalid state. So, SR-flip-flop is not suitable for an input S = R = 1.

Truth Table:

CLK	S	R	Q^+	State
0	X	X	Q	Hold
1	0	0	Q	Hold
1	0	1	0	Reset
1	1	0	1	Set
1	1	1	X	Invalid

Table 6.12

Characteristics Table:

CLK	S	R	Q	Q^+	State
1	0	0	0	0	Hold
1	0	0	1	1	
1	0	1	0	0	Reset
1	0	1	1	0	
1	1	0	0	1	Set
1	1	0	1	1	
1	1	1	0	x	Invalid
1	1	1	1	x	

Table 6.13

Characteristic Equation

$$Q^+(S,R,Q) = \sum m(1,4,5) + \sum d(6,7)$$

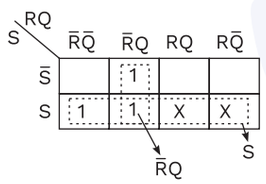


Fig. 6.19

$$Q^+(S,R,Q) = S + \bar{R}Q$$

Above equation is not valid for S = R = 1.

Excitation Table: Excitation table can be drawn using characteristic table.

Q	Q^+	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

Table 6.14 Excitation Table of SR Flip-flop

- If past output (Q) is logic '0' and present output (Q^+) is also logic '0', then S=logic '0' and R=don't care (logic '1' / logic '0').
- If past output (Q) is logic '0' and present output (Q^+) is logic '1', then S=logic '1' and R= logic '0'.
- If past output (Q) is logic '1' and present output (Q^+) is logic '0', then S=logic '0' and R=logic '1'.
- If past output (Q) is logic '1' and present output (Q+) is also logic '1', then S=don't care (logic '1' / logic '0') and R=logic '0'.

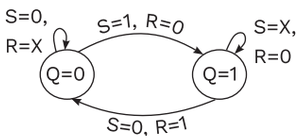


Fig. 6.20 State Diagram of SR Flip Flop

Previous Years' Questions

The circuit given below is a:

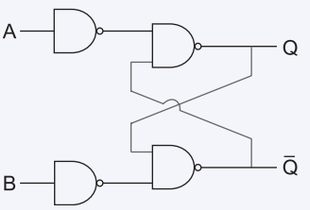


Fig. 6.21

(A) J-K Flip-flop
(B) Johnson's counter
(C) R-S latch
(D) None of the above
Solution: (C)

Drawbacks

In Sr flip-flop, due to the presence of an invalid state, there is a restriction on the sequence of the applied input, since it leads to CRITICAL RACE, which is undesirable.

J-K Flip-Flop

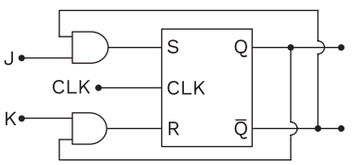


Fig. 6.22 J K Flip-flop

J-K flip-flop is the modified form of SR flip-flop, which overcome the problem of SR-flipflop (at S = R = 1, invalid state occurs) we use the feedback unit from the terminal Q and $\bar{Q}$ to an input of NAND gate.

$S = J\overline{Q}$ and $R = KQ$

The characteristic equation of SR flip-flop

$$Q^+ = S + \overline{R}Q$$

$$Q^+ = J\overline{Q} + \overline{KQ}\,Q$$

$$Q^+ = J\overline{Q} + \overline{K}\,Q$$

It is the characteristics equation of J-K flip-flop (modified SR flip-flop)

Case 1: For J = K = 0,

$$Q^+ = 0 \cdot \overline{Q} + 1 \cdot Q$$

and Q+ = Q

Case 2: For J = 0 and K = 1,

Q+ = 0 + 0

Q+ = 0

Case 3: For J = 1 and K = 0

$$Q^+ = 1 \cdot \overline{Q} + 1 \cdot Q$$

Q+ = 1

Case 4: For J = K = 1,

$$Q^+ = 1 \cdot \overline{Q} + 0$$

$$Q^+ = \overline{Q}$$

Previous Years' Questions

An S-R latch is implemented using TTL gates as shown in the figure. The set and reset pulse inputs are provided using the push-button switches. It is observed that the circuit fails to work as desired. The S-R latch can be made functional by changing

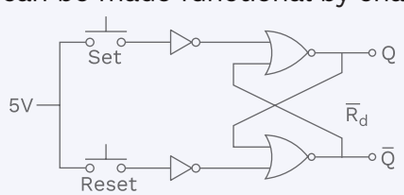


Fig. 6.23

(A) NOR gates to NAND gates
(B) Inverters to buffers
(C) NOR gates to NAND gates and inverters to buffers
(D) 5V to ground

Solution: (D)

Truth Table:

CLK	J	K	Q^+	State
0	×	×	Q	Hold
1	0	0	Q	Hold
1	0	1	0	Reset
1	1	0	1	Set
1	1	1	$\overline{Q}$	Toggle

Table 6.15 Truth Table of JK Flip-flop

Characteristic Table:

CLK	J	K	Q	Q^+	State
1	0	0	0	0	Hold
1	0	0	1	1	
1	0	1	0	0	Reset
1	0	1	1	0	
1	1	0	0	1	Set
1	1	0	1	1	
1	1	1	0	1	Toggle
1	1	1	1	0	

Table 6.16 Characteristic of JK Flip-flop

Characteristic equation:

Q+ (J, K, Q) = ∑m(1, 4, 5, 6)

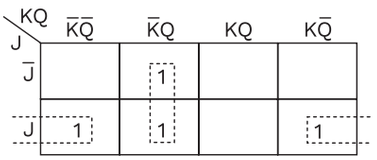


Fig. 6.24

$$Q^+(J,K,Q) = J\overline{Q} + \overline{K}Q$$

Previous Years' Questions

In figure A = 1 and B = 1, the input B is now replaced by a sequence 1010100 ....., the output X and Y will be:
(A) Fixed at 0 and 1 respectively
(B) X = 1010 ..... while Y = 0101 .....
(C) X = 1010 ..... while Y = 1010 .....
(D) Fixed at 1 and 0 respectively
Solution: (A)

Excitation Table:

Q	Q^+	J	K
0	0	0	×
0	1	1	×
1	0	×	1
1	1	×	0

Table 6.17 Excitation Table JK Flip-flop

- If past output (Q) is logic '0' and present output (Q^+) is also logic '0', then J = logic '0' and K = don't care.
- If past output (Q) is logic '0' and present output (Q^+) is logic '1', then J = logic '1' and K = don't care.
- If past output (Q) is logic '1' and present output (Q^+) is logic '0', then J = don't care and K = logic '1'.
- If past output (Q) is logic '1' and present output (Q^+) is also logic '1', then J = don't care and K = logic '0'.

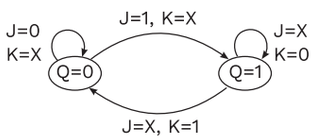


Fig. 6.25 State Diagram of JK Flipflop

D Flip-Flop:

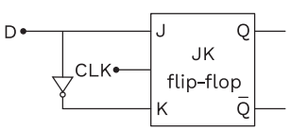


Fig. 6.26 D Flip-flop

- In D-flip-flop, terminal J of J-K flip-flop connected to input D and the terminal K of J-K flip-flop connected to a complement of an input $\bar{D}$.

Characteristic equation of J-K flip-flop:

$$Q^+ = J\bar{Q} + \bar{K}Q$$

$$Q^+ = D\bar{Q} + DQ$$

$$Q+ = D$$

- In D flip-flop, the present output (Q+) is equals to present input (D), So we use D flip-flop to provide delay in the circuit.

Truth Table:

CLK	D	Q^+
0	×	Q
1	0	0
1	1	1

Table 6.18 Truth Table of D Flip-flop

Characteristic Table:

CLK	D	Q	Q^+	State
0	×	Q	Q	Hold
1	0	0	0	Reset
1	0	1	0	
1	1	0	1	Set
1	1	1	1	

Table 6.19 Characteristic Table of D Flip-flop

Characteristic equation,

$$Q+ (D, Q) = \sum m(2, 3)$$

$$Q^+(D, Q) = D\bar{Q} + DQ$$

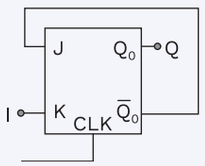
$$Q+ (D, Q) = D$$

Present output = Present input

- D flip-flop is used as a delay and storage element e.g.: memory devices, registers.

Previous Years' Questions

Consider the following J-K flip-flop in the above J-K flip-flop, J = Q and K = 1. Assume that the flip-flop was initially cleared and then clocked for 6 pulses. What is the sequence at the Q output?



(A) 010000 (B) 011001
(C) 010010 (D) 010101
Solution: (D)

Excitation Table:

Q	Q^+	D
0	0	0
0	1	1
1	0	0
1	1	1

Table 6.20 Excitation Table of D Flip-flop

- If past output (Q) is logic '0' and present output (Q^+) is also logic '0', then D = Q^+ = logic '0'.
- If past output (Q) is logic '0' and present output (Q^+) is logic '1', then D = Q^+ = logic '1'.
- If past output (Q) is logic '1' and present output (Q^+) is logic '0', then D = Q^+, then D = Q^+ = logic '0'.

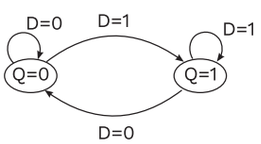


Fig. 6.27 State Diagram of D Flip-flop

- If past output (Q) is logic '1' and present outout (Q^+) is also logic '1', then D = Q^+ = logic '1'.
- D flip-flop and JK-flip-flop can be replaced by the SR-flip-flop, because it can overcome invalid state (S = R = 1) problem of the SR flip-flop.



Fig. 6.19 SR Flip-flop in Term of D Flip-flop

T Flip-Flop: (Toggle Flip-flop)

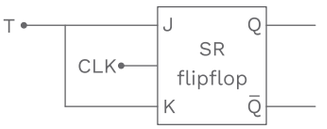


Fig. 6.28 T Flip-flop

- In T-flip-flop both the terminal of J-K flip-flop connected to the input T(shorting the input terminal of J-K flip-flop).

Characteristic equation of JK-flip-flop,

$$Q^+ = J\overline{Q} + \overline{K}Q$$

$$Q^+ = T\overline{Q} + \overline{T}Q$$

$$Q+ = T \oplus Q$$

Truth Table:

CLK	T	Q^+
0	×	Q
1	0	Q
1	1	$\overline{Q}$

Table 6.21 Truth Table of T Flip-flop

Characteristic Table:

CLK	T	Q	Q^+	State
0	×	Q	Q	Hold
1	0	0	0	Hold
1	0	1	1	

CLK	T	Q	Q^+	State
1	1	0	1	Toggle
1	1	1	0	

Table 6.22 Characteristic Table of T Flip-flop

Characteristic Equation,

$$Q+(T, Q) = \Sigma m(1, 2)$$

$$Q^+ = \overline{T}Q + T\overline{Q}$$

$$Q+ = T \oplus Q$$

Excitation Table:

CLK	T	Q^+
0	0	0
0	1	1
1	0	1
1	1	0

Table 6.23 Excitation Table of T Flip-flop

Previous Years' Questions

A sequential circuit using D Flip-flop and logic gates is shown in figure, where X and Y are the inputs and Z is the output. The circuit is

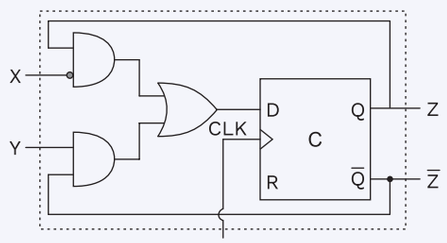


Fig. 6.29

(A) S-R flip-flop with inputs X = R and Y = S

(B) S-R flip-flop with inputs X = S and Y = R

(C) J-K flip-flop with inputs X = J and Y = K

(D) J-K flip-flop with inputs X = K and Y = J

Solution: (D)

- If the past output (Q) is logic '0' and the present output (Q^+) is also logic '0', then T = logic '0'.
- If the past output (Q) is logic '0' and the present output (Q^+) is logic '1', then T = logic '1'.
- If the past output (Q) is logic '1' and the present output (Q^+) is logic '0', then T = logic '1'.
- If the past output (Q) is logic '1' and the present output (Q^+) is also logic '1', then T = logic '0'.

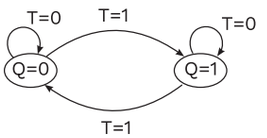


Fig. 6.30

Triggering:

- The momentary change in the control input of a flip-flop to switch it from one state to another state is called a trigger, and the transition that causes is said to trigger the flip-flop. The process of applying the control signal to change the stage of the flip-flop is called triggering. There are two types of triggering in the flip-flop.

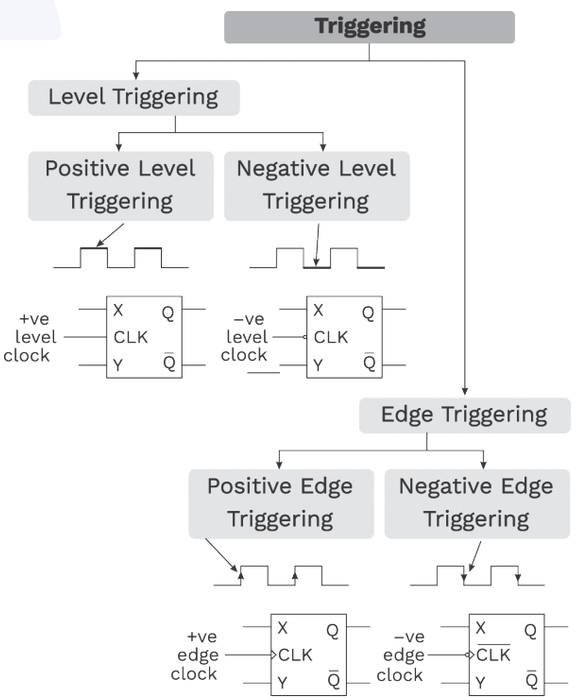


Fig. 6.31 Type of Triggering

- Level triggering is used to trigger the flip-flop for the some interval of time (t_1 – t_2).
- Edge triggering is used to trigger the flip-flop at the instant of time (t = t_1).

Example 5:

Draw the output waveform for the given input of SR-flip-flop having a positive level trigger clock.

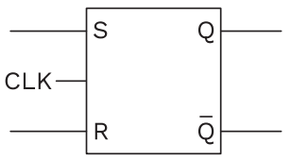


Fig. 6.32

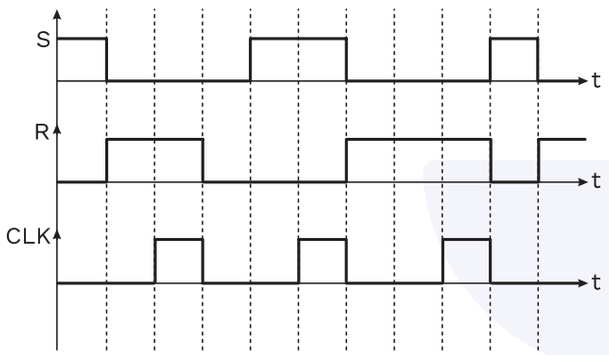


Fig. 6.33 Positive Level Trigger

Solution:

Given, flip-flop triggers when the positive level will appear in a clock signal.

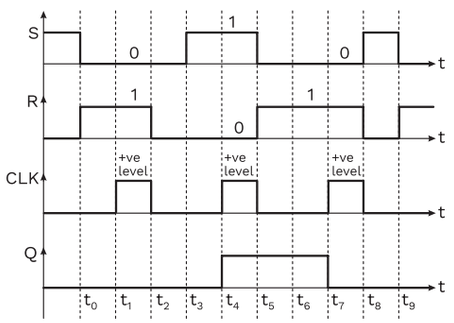
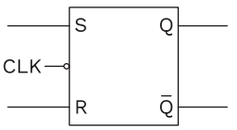


Fig. 6.34 Positive Level Trigger

Example 6:

Draw the output waveform for the given input of SR-flip-flop having a negative level trigger clock.



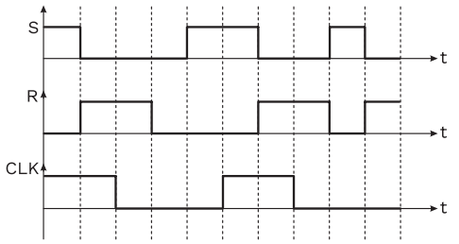


Fig. 6.35 Negative Level Trigger

Solution:

Given flip-flop triggers when the negative level appear in a clock signal.

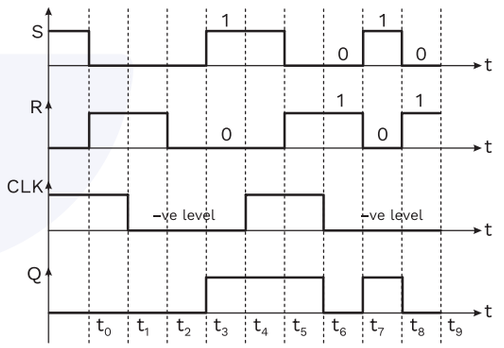
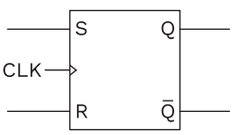


Fig. 6.36 Negative Level Trigger

Example 7:

Draw the output waveform for the given input of SR-flip-flop having a positive edge trigger clock.



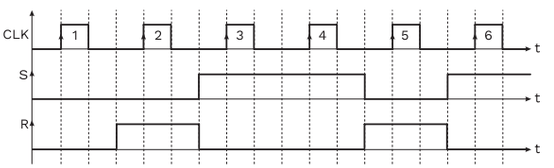


Fig. 6.37 Positive Edge Trigger

Solution:

Given flip-flop triggers when the positive edge will appear in a clock signal.

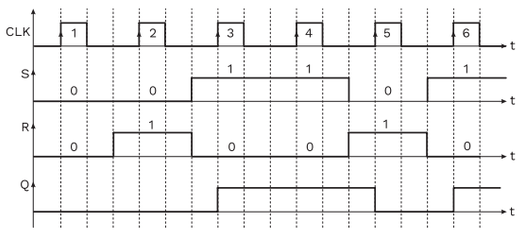


Fig. 6.38 Positive Edge Trigger

Conversion of flip-flop:

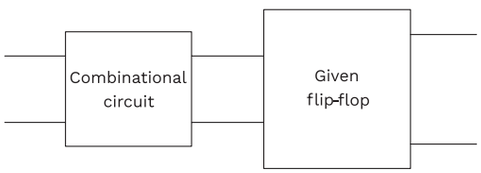


Fig. 6.39 Conversion of Flip-flop

Previous Years' Questions

The present output Qn, of an edge triggered J-K flip-flop is logic 0. If J = 1, then Q_{n+1}

(A) Cannot be determined
(B) Will be logic 0
(C) Will be logic 1
(D) Will race around

Solution: (C)

In order to convert one type of flip-flop to the another type of flip-flop, follow below steps for conversion.

Step 1: Write the state table of the required flip-flop.

Step 2: Match the excitation table of the given flip-flop to the required flip-flop.

Step 3: Write the excitation expression.

Step 4: Minimize logical expression.

Step 5: Implement a logic circuit.

Example 8:

Convert SR-flip-flop to JK-flip-flop.

Solution:

Given flip-flop = SR-flip-flop

Required flip-flop = J-K flip-flop

Step 1: Characteristic table/state table of JK-flip-flop,

J	K	Q	Q^+
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

Table 6.24

Step 2: Match the characteristic table of JK flip-flop with excitation table of SR-flip-flop

Characteristic Table of JK-flip-flop

J	K	Q	Q^+	S	R
0	0	0	0	0	×
0	0	1	1	×	0
0	1	0	0	0	×
0	1	1	0	0	1
1	0	0	1	1	0
1	0	1	1	×	0
1	1	0	1	1	0
1	1	1	0	0	1

Excitation table of SR-flip-flop

Table 6.25

Step 3: Excitation expression:

$S(J, K, Q) = \sum m(4, 6) + d(1, 5)$

$R(J, K, Q) = \sum m(3, 7) + d(0, 2)$

Step 4: Minimize logical expression using K-map.

K-map for S,

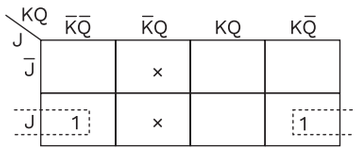


Fig. 6.40

$S(J,K,Q) = J\bar{Q}$

K-map for R,

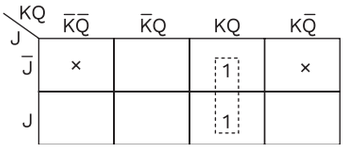


Fig. 6.42

$R(J, K, Q) = KQ$

Step 5: Logic circuit,

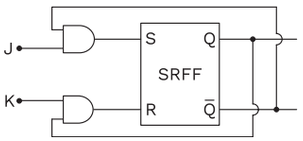


Fig. 6.43

Previous Years' Questions

The circuit shown in the figure is a:

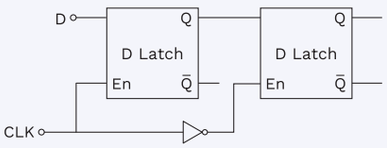


Fig. 6.41

(A) Toggle Flip Flop
(B) J-K Flip Flop
(C) SR Latch
(D) Master-Salve D flip Flop

Solution: (D)

Example 9:

Convert the SR-flip-flop to the XY-flip-flop, whose truth table is given below:

X	Y	Q^+
0	0	1
0	1	$\bar{Q}$
1	0	Q
1	1	0

Table 6.26

Solution:

To convert SR-flip-flop to XY-flip-flop, characteristic table of XY-flip-flop and excitation table of SR-flip-flop required.

Characteristic Table of XY-flip-flop

Excitation table of SR-flip-flop

X	Y	Q	Q^+	S	R
0	0	0	1	1	0
0	0	1	1	×	0
0	1	0	1	1	0
0	1	1	0	0	1
1	0	0	0	0	×
1	0	1	1	×	0
1	1	0	0	0	×
1	1	1	0	0	1

Table 6.27

$$S(X, Y, Q) = \sum m(0, 2) + \sum d(1, 5)$$
$$R(X, Y, Q) = \sum m(3, 7) + \sum d(4, 6)$$

Previous Years' Questions

An S-R FLIP-FLOP can be converted into a T FLIP-FLOP by connecting _____ to Q and _____ to $\overline{Q}$.

Solution: R and S

K-map of S,

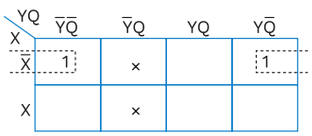


Fig. 6.44

$S(X, Y, Q) = \overline{X}\overline{Q}$

K-map of R,

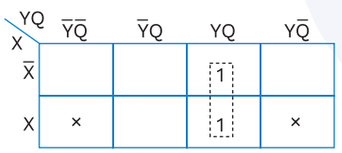


Fig. 6.45

$R(X, Y, Q) = YQ$

Logic circuit

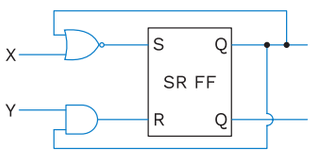


Fig. 6.46

Conversion Table of Flip-Flops

FF-1 to FF-2	Expression
JK-FF to SR-FF	J = S and K = R
JK-FF to D-FF	J = D and $K = \overline{D}$
JK - FF to T-FF	J = T and K = T
SR-FF to JK-FF	$S = J\overline{Q}$ and R = KQ
SR-FF to D-FF	S = D and $R = \overline{D}$
SR-FF to T-FF	$S = T\overline{Q}$ and R = TQ
D-FF to SR-FF	$D = S + \overline{R}Q$
D-FF to JK-FF	$D = J\overline{Q} + \overline{K}Q$
D-FF to T-FF	$D = T \oplus Q$
T-FF to SR-FF	$T = S\overline{Q} + RQ$
T-FF to JK-FF	$T = J\overline{Q} + KQ$
T-FF to D-FF	$T = D \oplus Q$

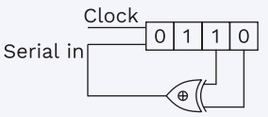
Table 6.28 Conversion Table of Flip-flop

Applications of flip-flops:

Some of the applications of flip-flops are:
1. Latch
2. Serial to parallel data storage
3. Serial to parallel converter
4. Parallel to serial converter
5. Frequency division
6. Data transfer
7. Counters

Previous Years' Questions

The initial contents of the 4-bit serial-in parallel-out, right-shift, Shift Register shown in the figure is 0110. After three C pulses are applied, the contents or the Shift Register will be:



(A) 0000 (B) 0101
(C) 1010 (D) 1111

Solution: (C)

Shift Registers

1. Flip-flop is a single bit memory element. Which can't store multiple bits at a time, so we combine desired number of flip-flops, the resultant circuit, which stores multiple bits, is known as registers.
2. Since no data manipulation is required, so we prefer D-flip-flops for this purpose. For D-flipflop

   $Q+ = D$

3. To store n-bits, n-flip-flops are required. The data is available in two forms,
   a) Serial data (Temporal code) → Depends on time
   b) Parallel data (Spatial code) → Depends on space

4. Depending on input and output registers are classified into 4 types.
   a) Serial In Serial Out (SISO)
   b) Serial In Parallel Out (SIPO)
   c) Parallel in Parallel Out (PIPO)
   d) Parallel In Serial Out (PISO)

1. Serial In serial out (SISO):

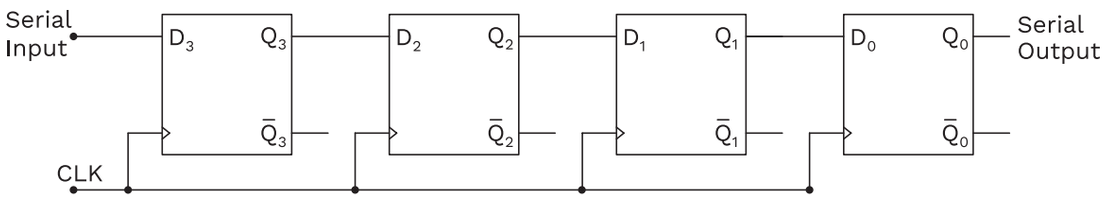


Fig. 6.47

a) SISO register configuration accepts serial data input and produces serial data output.
b) It accepts one bit at a time on a single input line.
c) Block diagram Representation,

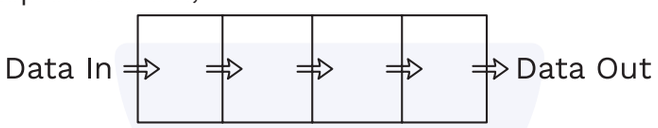


Fig. 6.48

d) SISO configuration has only 1-input and 1-output.
e) For SISO configuration,
   Storing n-bit data = n-clock pulses required
   Retrieving n-bit data = (n-1) clock pulses required total number of clock pulses = (2n – 1) clock pulses
   i) Let serial input data is 1010.

CLK	Input	Q_3	Q_2	Q_1	Q_0
0	×	0	0	0	0
1	0	0	0	0	0
2	1	1	0	0	0
3	0	0	1	0	0
4	1	1	0	1	0
5	×	×	1	0	1
6	×	×	×	1	0
7	×	×	×	×	1

For storing 4-bit, 4-clock pulses required

Table 6.29 SISO Configuration

Previous Years' Questions

Assuming that all flip-flops are in reset condition initially, the count sequence observed at Q_A in the circuit shown is:

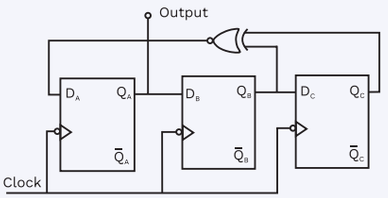


Fig. 6.49

(A) 0010111... (B) 0001011...
(C) 0101111... (D) 0110100...

Solution: (D)

2. Serial in parallel out (SIPO):

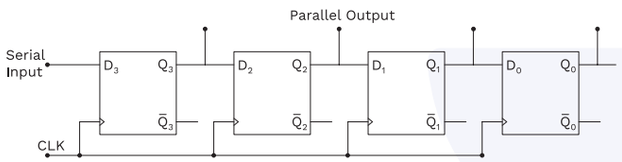


Fig. 6.50

a) SIPO register configuration accepts serial data input and produces parallel data output.

b) Block diagram Representation,

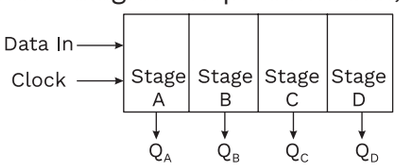


Fig. 6.51

c) SIPO configuration has only 1-input and multiple outputs.

d) For SIPO configuration,

Storing n-bit data = n-clock pulses required

Retrieving n-bit data = 0 (zero) clock pulse required

Total number of clock pulses = n – clock pulses

3. Parallel in parallel out (PIPO):

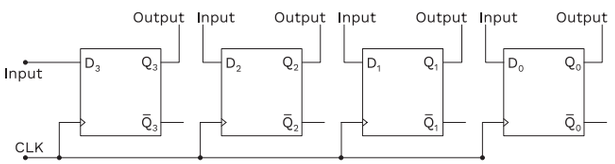


Fig. 6.52

a) PIPO register configuration accepts parallel data input and produces parallel data output.

b) Block diagram representation,

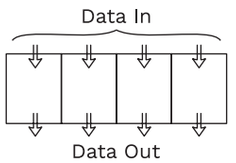


Fig. 6.53

c) PIPO configuration has multiple inputs and multiple outputs.

d) For PIPO configuration,

Storing n-bit data = 1 (one) clock pulse required

Retrieving n-bit = 0 (zero) clock pulse required

Total number of clock pulses = 1 clock pulse required

4. Parallel in serial out (PISO):

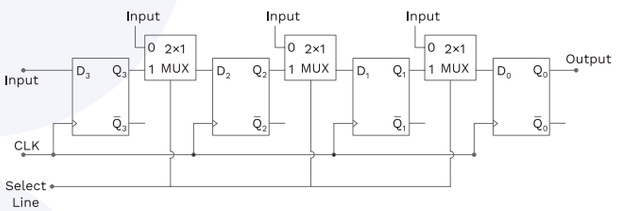


Fig. 6.54

a) PISO register configuration accepts parallel data input and produces serial data output.

b) Block diagram representation,

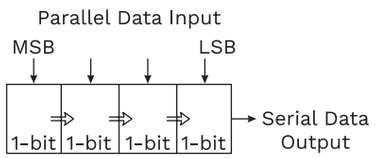


Fig. 6.55

c) PISO configuration has multiple inputs and 1-output.

d) For PISO configuration,

Storing n-bit data = 1 clock pulse required

Retrieving n-bit data=(n-1) clock pulses required

Total number of clock pulses = n-clock pulses required

Configuration	Clock Pulses
Serial Input	n-clock pulses required
Parallel Input	1-clock pulse required
Serial Output	(n-1) clock pulses required
Parallel Output	0-clock pulse required

Table 6.30 Shift Register and Clock Pulses

Example 10:

A 4-bit shift register circuit configured for right shift operation Din→ A, is shown. If the present state of the shift register is ABCD = 1001, the number of clock cycles required to reach the state ABCD = 1110 is _____.

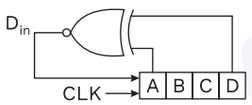


Fig. 6.56

Solution:

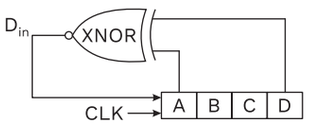


Fig. 6.57

For right shift SIPO shift register,

Input A = D_{in} = A ⊙ D

State Table:

CLK	Input	A	B	C	D
	A ⊙ D				
0	–	1	0	0	1
1	1	1	1	0	0
2	0	0	1	1	0
3	1	1	0	1	1
4	1	1	1	0	1
5	1	1	1	1	0

Table 6.31

Hence, the number of clocks required to reach the state ABCD = 1110 is 5.

Previous Years' Questions

For the circuit shown in the figures below, two 4-bit parallel-in serial-out shift registers loaded with the data mentioned are used to feed the data to a full adder. Initially, all the flip-flops were in clear state. After applying two clock pulses, the outputs of the full-adder should be:

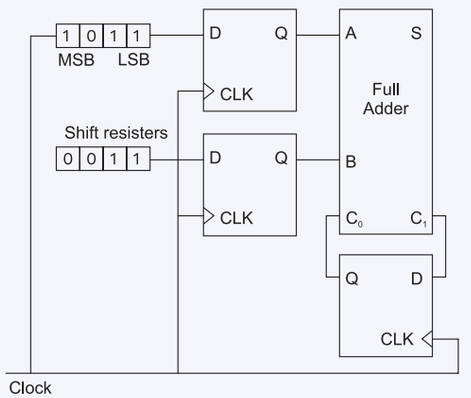


Fig. 6.58

(A) X = 0, C0 = 0
(B) X = 0, C0 = 1
(C) X = 1, C0 = 0
(D) X = 1, C0 = 1

Solution: (D)

Example 11:

An 8-bit register is made of one 4-bit PISO register (synchronous loading) cascaded with a 4-bit SIPO register, as shown in the figure below, the total number of clock pulses required to perform write and read operations for one byte is _____.

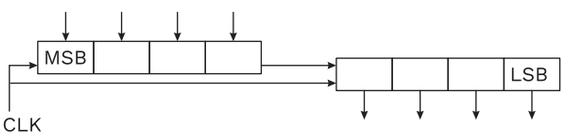


Fig. 6.59

Solution:

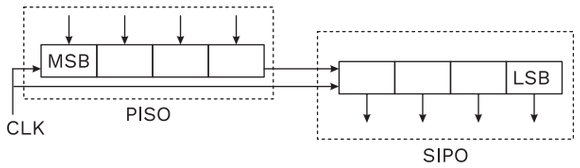


Fig. 6.60

Let 1-byte data is $D_7\ D_6\ D_5\ D_4\ D_3\ D_2\ D_1\ D_0$.

Write operation:

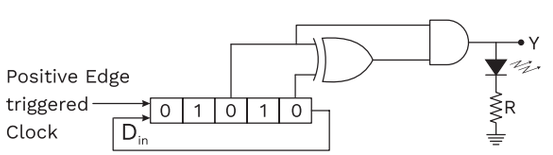
- To write the 4-bit data ($D_3\ D_2\ D_1\ D_0$) in PISO, we required a 1-clock pulse.
- To shift the 4-bit data ($D_3\ D_2\ D_1\ D_0$) from PISO to SIPO, we required 4-clock pulses (each bit need 1-clock pulse because in SIPO, we can write only 1-bit at every clock pulse).
- To write the 4-bit data ($D_7\ D_6\ D_5\ D_4$) again in PISO, we required a 1-clock pulse.

Read operation:

- To read the 4-bit data ($D_3\ D_2\ D_1\ D_0$) in SIPO, we required zero (0) clock pulse.
- To read the 4-bit data ($D_7\ D_6\ D_5\ D_4$) in PISO, we need to shift this data to SIPO, for shifting this data ($D_7\ D_6\ D_5\ D_4$) to SIPO 4-clock pulses required.
- To read the 4-bit data ($D_7\ D_6\ D_5\ D_4$) in SIPO, we required a zero (0) clock pulse.
- Total number of clock pulses required to read and write the 1-byte data ($D_7\ D_6\ D_5\ D_4\ D_3\ D_2\ D_1\ D_0$) is 10.

Example 12:

Consider a serial-in-parallel-out (SIPO) right shift register circuit with initial contents as '01010' as shown in the figure. The circuit is operated with a positive edge-triggered clock signal as given in the figure. If +5V and 0V are used to represent logic-1 and logic-0 respectively, then at which of the following instances the LED will be in the ON state?



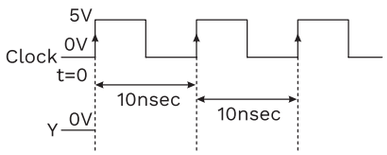


Fig. 6.61

(A) 15 nsec (B) 25 nsec
(C) 35 nsec (D) 45 nsec

Solution:

LED will glow when the output of AND gate become logic '1' (5V). For that output of XOR gate must be logic '1' and the input of XOR gate must be 1 and 0.

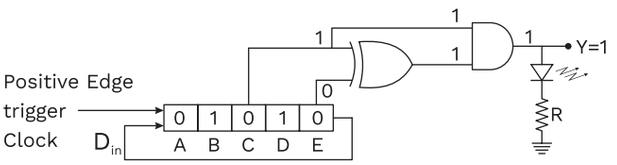


Fig. 6.62

So, at C logic '1' should appear and at E logic '0' should appear to glow the LED.

CLK	Time (nsec)	A	B	C	D	E
Initially 0		0	1	0	1	0
1	0–10	0	0	1	0	1
2	10-20	1	0	0	1	0
3	20–30	0	1	0	0	1
4	30–40	1	0	1	0	0

Table 6.32

Therefore, LED will glow between 30 nsec to 40 nsec.

Hence, option 'C' is correct.

Toggle Modes

SR-flip-flop = 1 Toggle Mode

JK-flip-flop = 4 Toggle Mode

D-flip-flop = 1 Toggle Mode

T-Flip-flop = 1 Toggle Mode

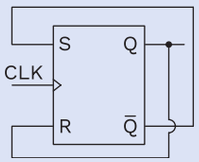
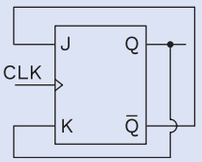
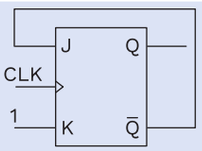
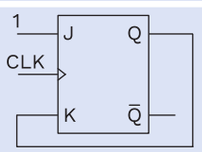
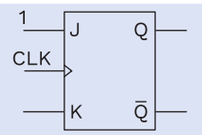
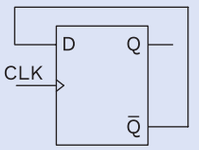
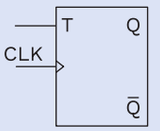
SR-Flip-Flop	JK-Flip-Flop	D-Flip-Flop	T-Flip-Flop
 For Toggle mode, $T_{out} = 2T_{CLK}$ $f_{out} = \frac{f_{CLK}}{2}$	    For All Toggle Mode $T_{out} = 2T_{CLK}$ $f_{out} = f_{CLK}/2$	 For Toggle mode, $T_{out} = 2T_{CLK}$ $f_{out} = \frac{f_{CLK}}{2}$	 For Toggle mode $T_{out} = 2T_{CLK}$ $f_{out} = \frac{f_{CLK}}{2}$

Table 6.33 Toggle Modes

Example 13:

The frequency of the clock applied to the rising edge-triggered D flip-flop shown in the figure is 20kHz. The frequency of the signal available at Q is _____.

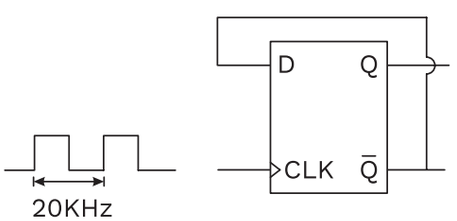


Fig. 6.63

(A) 10 kHz (B) 5 kHz
(C) 20 kHz (D) 40 kHz

Solution:

The given, D flip-flop is in Toggle mode.

$$f_{out} = \frac{f_{CLK}}{2}$$

$$f_{out} = \frac{20}{2}$$

fout = 10 kHz

Correct option is (A).

Example 14:

Find the frequency of Q0 and Q1, shown in the figure, if the clock frequency is 40 kHz.

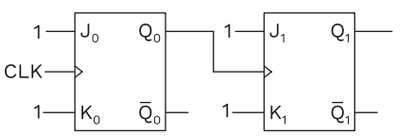


Fig. 6.64

Solution:

Given, flip-flop is in Toggle mode,

$$f_{Q_0} = \frac{f_{CLK}}{2}$$

$$f_{Q_0} = \frac{40}{2}$$

$$f_{Q_0} = 20\,kHz$$

The frequency of output Q0 is the clock frequency for the 2nd JK-flip-flop.

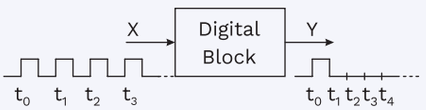
$$f_{Q_1} = \frac{f_{Q_0}}{2}$$

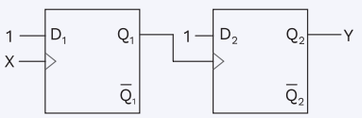
$$f_{Q_1} = \frac{20}{2}$$

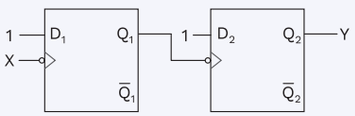
$$f_{Q_1} = 10\ kHz$$

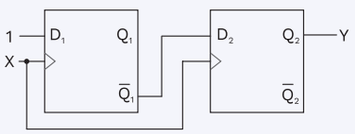
Previous Years' Questions

The digital block in the figure is realized using two positive edge triggered D-flip-fops. Assume that t<t_0, $Q_1 = Q_2 = 0$. The circuit int eh digital block is given by:



(A)


(B)


(C)


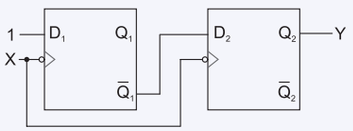
(D)


Fig. 6.65

Solution: (C)

Example 15:

In a JK flip-flop we have $J = \overline{Q}$ and K = 1. Assuming that flip-flop was initially cleared, and then clocked for 5 pulses, the sequence at the Q output will be:

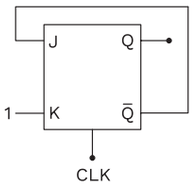


Fig. 6.66

(A) 01101 (B) 01010
(C) 10010 (D) 11100

Solution:

Given, J-K flip-flop is in Toggle mode. Hence, the output Q = 01010 for the 5 clock pulses. The correct option is (B).

Example 16:

Find the frequency of F and duty cycle of output, if the duty cycle of the clock is 50%.

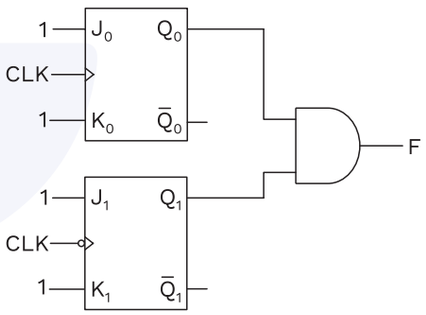


Fig. 6.67

Solution:

Both flip-flop are in Toggle mode.

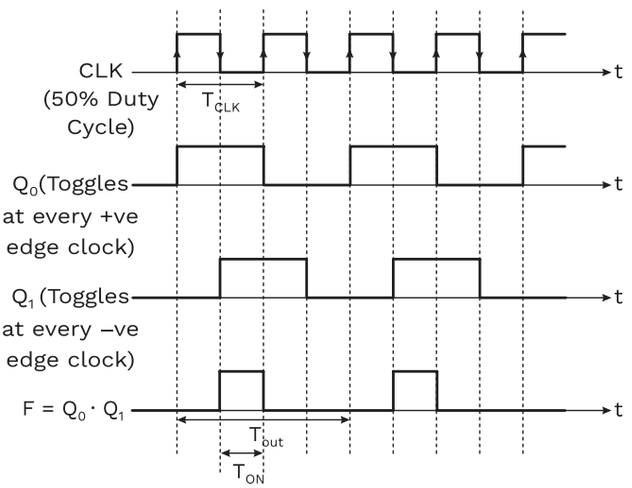


Fig. 6.68

Toutput = 2TCLK

$$f_{output} = \frac{f_{CLK}}{2}$$

$$\text{Duty cycle (D)} = \frac{T_{ON}}{T_{ON} + T_{OFF}} \times 100\%$$

$$D = \frac{1}{4} \times 100\%$$

$$D = 25\%$$

- **Race around condition:**

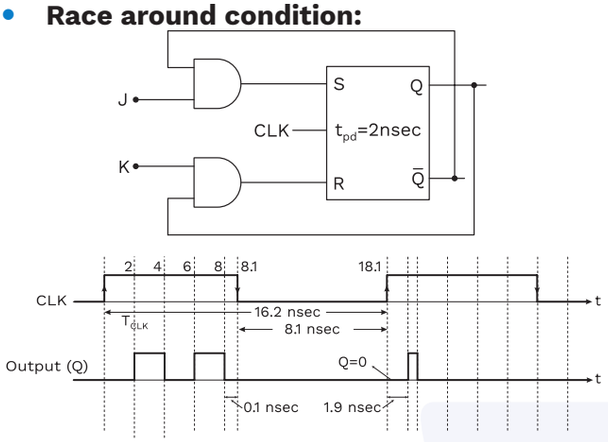


Fig. 6.69

Case 1: Let J = 1, K = 1, T_{CLK} = 16.2 nsec and propagation delay of SR flip-flop t_{pd} = 2nsec.

Assuming SR-flip-flop is initially reset (Q=0 and $\bar{Q} = 1$) so the input of SR-flip-flop will be S = 1 and R = 0 for the first positive level clock and output (Q) will generate after 2 nsec due to propagation delay of SR-flip-flop.

- For S = 1 and R = 0, output (Q) generated as logic '1' (Q = 1 and $\bar{Q} = 0$), at the same instant input of SR flip-flop will changes to S = 0 and R = 1.
- For S = 0 and R = 1, output (Q) generated as logic '0' (Q = 0 and $\bar{Q} = 1$), at the same instant input of SR-flip-flop will change to S = 1 and R = 0.
- For S = 1 and R = 0, output (Q) required 2 nsec (tpd) to generate the next output (Q) as logic '1', but after 0.1 nsec clock changes its state and SR-flip-flop output will wait for the next positive level clock pulse, after the whole cycle of TCLK (16.2 nsec), the output of SR flip-flop need 1.9 nsec more to generate the output Q = 1.

Case 2: Let J = 1, K = 1, T_{CLK} = 20.2 nsec and propagation delay of SR flip-flop t_{pd} = 2nsec.

Assuming SR-flip-flop is initially reset (Q = 0 and $\bar{Q} = 1$) so the input of SR-flip-flop will be S = 1 and R = 0, for the first positive level clock and output (Q) will generate after 2 nsec due to propagation delay of SR-flip-flop.

- For S = 1 and R = 0, output (Q) generated as logic '1' (Q=1 and $\bar{Q} = 0$), at the same instant input of SR flip-flop will changes to S = 0 and R = 1.
- For S = 0 and R = 1, output (Q) generated as logic '0' (Q = 0 and $\bar{Q} = 1$), at the same instant input of SR-flip-flop will changes to S = 1 and R = 0.
- For S = 1 and R = 0, output (Q) required 2 nsec (tpd) to generate the next output (Q) as logic '1', but after 0.1 nsec clock changes its state and SR-flip flop output will wait for the next positive level clock pulse, after whole cycle of TCLK (20.2 nsec), the output of SR flip-flop will be Q = 1 till 22.1 nsec.

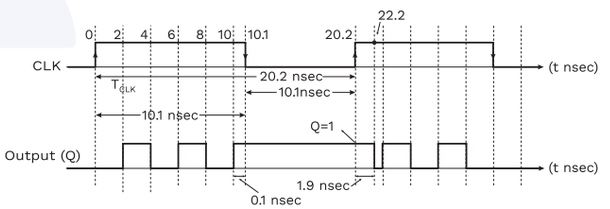


Fig. 6.70

In case (1) when we apply the 2nd positive level clock pulse, at that instant of time output was Q = 0.

In case (2) when we apply the 2nd positive level clock pulse, at that instant of time output was Q = 1.

Previous Years' Questions

Consider the given circuit.

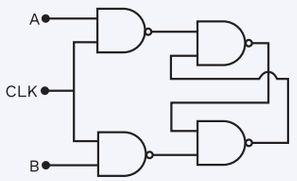


Fig. 6.71

In this circuit, the race around:

(A) Does not occur
(B) Occurs when CLK = 0
(C) Occurs when CLK=1 and A=B=1
(D) Occurs when CLK=1 and A=B=0

Solution: (A)

So, the output of the flip-flop changes to 0 → 1 → 0 .... continuously the starting of the next clock, which makes the output uncertain, it is called as Race Around Condition (RAC).

1. Race around condition (RAC) could occur in any flip-flop if the following conditions satisfies.
   a) If the flip-flops are operated in level triggering
   b) If, $t_{pd} < (T_{CLK})_{ON}$
   c) If the flip-flops are operated in toggle mode.
2. If the above conditions satisfy simultaneously, then there is a continuous race in the output of the flip-flop between 0 and 1 to reach the next state, who will be the winner of this race is not certain, that depends on tpd and $(T_{CLK})_{ON}$.
3. Remedy:
   a) (TCLK)ON < (tpd) < T
   b) By using edge-triggered flip-flop.
   c) By Master-slave configuration.

Master-Slave Configuration

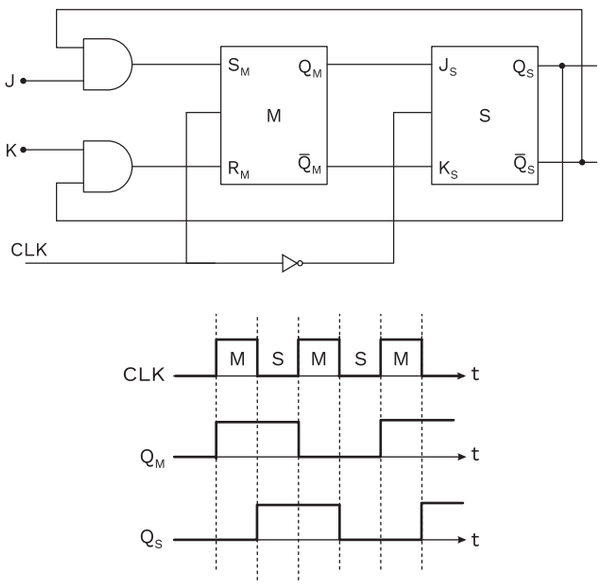


Fig. 6.72

In the case of Master-slave configuration, the master is applied with input clock and slave is applied with inverted clock, for input J = 1 and K = 1, out of two flip-flops at a time, only one of the flop-flop respond and other will not respond. As a result, many times toggling in a single clock cycle has been converted to only one-time toggle, hence Race around condition is avoided.

Previous Years' Questions

The ripple counter shown in the given figure works as a:

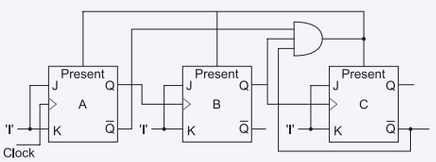


Fig. 6.73

(A) MOD-3 up counter
(B) MOD-5 up counter
(C) MOD-3 down counter
(D) MOD-5 down counter

Solution: (D)

- In master-slave configuration, the command signal is generated by master flip-flop and the response of the command signal is given by slave flip-flop.
- Master-slave flip-flop can store only 1-bit of data.

Counters

Counters are the combination of various flip-flops, that generate a desired counting patterns when the clocks are applied.

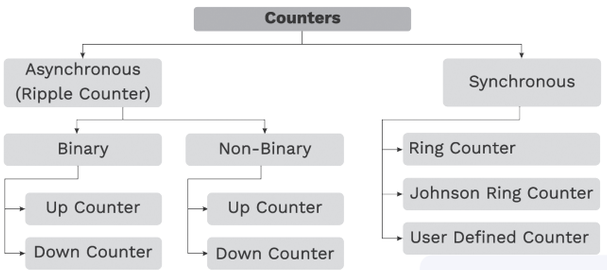


Fig. 6.74

- In synchronous counter, all flip-flops respond simultaneously.
- In asynchronous counter, every flip-flop requires a different clock and flip-flop response depends on when the clock is applied.
- Any possible output of a counter is known as its state; for a n-bit counter the maximum possible states are 2n.
- The states which are counted by the counter are called valid states. i.e., if we required only 0, 2, 3, 5, 7 states on any counter then these states are called valid states.
- The states which are not counted (skipped) by the counter are called as invalid states. So, the invalid states will be 1, 4, 6.
- The minimum number of clocks needed to get the counting pattern repeats are called as modulus of a counter (the number of states in count sequence).
  If we have states as.

  $\underline{00 \rightarrow 01 \rightarrow 10 \rightarrow 11} \rightarrow 00 \rightarrow 01 \rightarrow 10 \rightarrow 11$

  Here, number of states in count sequence = 4.
  So, MODULUS of counter = 4.

Previous Years' Questions

In the modulo-6 ripple counter shown in below, figure the output of the 2-input gate is used to clear the J-K flip-flops

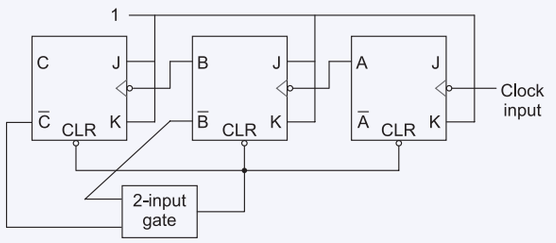


Fig. 6.75

The 2-input gate is:
(A) a NAND gate
(B) a NOR gate
(C) an OR gate
(D) an AND gate
Solution: (C)

Example 17:

What is the MOD number of the counting sequence?

00 → 00 → 10 → 10 → 01 → 01 → 11 → 11 → 00 → 00 → 10 → 10 → 01 → 01 → 11 → 11

Solution:

Given sequence,

00 → 00 → 10 → 10 → 01 → 01 → 11 → 11 → 00 → 00 → 10 → 10 → 01 → 01 → 11 → 11

States of sequence,

$\underline{0 \rightarrow 0 \rightarrow 2 \rightarrow 2 \rightarrow 1 \rightarrow 1 \rightarrow 3 \rightarrow 3} \rightarrow 0 \rightarrow 0 \rightarrow 2 \rightarrow$

$2 \rightarrow 1 \rightarrow 1 \rightarrow 3 \rightarrow 3$

The number of states in count sequence = 8, Hence the MOD of counter = 8.

Example 18:

What is the MOD number of the counting sequence-

00 → 01 → 00 → 10 → 00 → 11 → 00 → 01 → 00 → 10 → 00 → 11 → ....

Solution:

Given sequence,

00 → 01 → 00 → 10 → 00 → 11 → 00 → 01 → 00 → 10 → 00 → 11 → ....

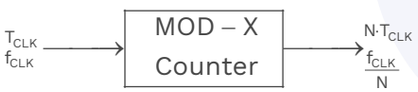
States of sequence,

$\underbrace{0 \rightarrow 1 \rightarrow 0 \rightarrow 2 \rightarrow 0 \rightarrow 3}$ → 0 → 1 → 0 → 2 → 0 → 3

Number of states in count sequence = 6, Hence the MOD of counter = 6.

- For n-flip-flops, the maximum number of possible state = 2^n.
- If N is MOD number of a counter, then
- $N \leq 2^n$; where n = number of flip-flops
- Time period of counter output (T_{out}) = (MOD) T_{CLK}

$$f_{out} = \frac{f_{CLK}}{MOD}$$



Example 19:

A counter has 1025 used states, for the designing of given counter, how many flip-flops are required?

Solution:

The counter has 1025 used states. So the MOD of the counter N = 1025.

Number of flip-flops required for MOD-1025 counter.

$N \leq 2n$

$1025 \leq 2n$

$n = 11$

Hence, the number of flip-flops required = 11

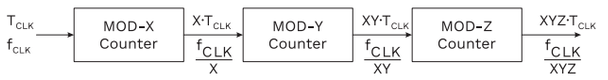


Fig. 6.76

Previous Years' Questions

The given figure shows a ripple counter using positive edge triggered Flip-lops. If the present state of the counter is $Q_2Q_1Q_0$ = 011 then its next state $Q_2Q_1Q_0$ will be:

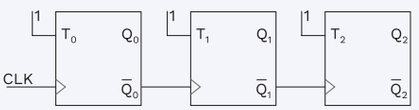


Fig. 6.77

(A) 010
(B) 100
(C) 111
(D) 101

Solution: (B)

Comparison of Asynchronous and Synchronous Counters

Asynchronous Counters	Synchronous Counters
Different flip-flops are applied with different clocks.	All flip-flops are applied with same clock.
Designing and implementation is very simple even for more number of states.	Design and implementation become complex as the numbers of states increases.
Slower compared to synchronous counters.	Faster compared to asynchronous counters.
Transition states are present.	Transition states are not present.

Asynchronous Counters	Synchronous Counters
Only fixed counting sequence is possible to implement. 1. UP counting 2. Down counting	Any counting sequence is possible.

Table 6.34 Comparison of Asynchronous and Synchronous Counters

Asynchronous Counter

1. Different flip-flops are applied with different clocks.
2. In asynchronous counter, the external clock is applied only at LSB and output of next flip-flop will act as a clock to the next flip-flops.
3. Flip-flops are operated in toggle mode.
4. Fixed counting sequence.
   a) UP Counter
   b) Down Counter

3-Bit Asynchronous Counter/3-Bit Ripple Counter

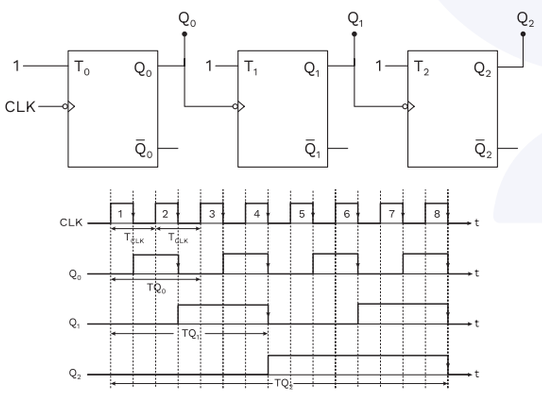


Fig. 6.78

So, TQ0 = 2TCLK

TQ1 = 4TCLK

TQ2 = 8TCLK

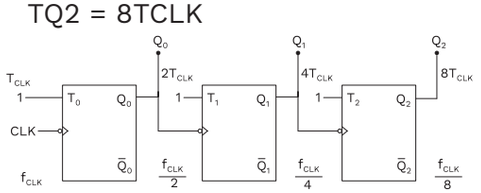


Fig. 6.79

- Output Q_0 will change (toggles) for every negative edge of the clock.
- Output Q_1 will change (toggled) for every negative edge of output Q_0 (Q_0 changes 1 → 0, Q_1 toggles).
- Output Q_2 will change (toggles) for every negative edge of output Q_1 (Q_1 changes 1 → 0, Q_2 toggles).

Previous Years' Questions

In the figure, the J and K inputs of all the four flip-flops are made high. The frequency of the signal at output Y is:

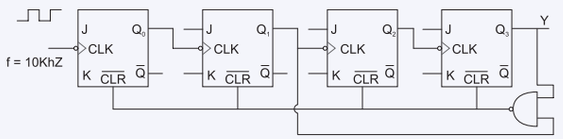


Fig. 6.80

(A) 0.833 kHz (B) 1.0 kHz
(C) 0.91 kHz (D) 0.77 kHz

Solution: (B)

	CLK	Q_2	Q_1	Q_0
t_1 →	0	0	0	0
	1	0	0	1
	2	0	1	0
	3	0	1	1
	4	1	0	0
	5	1	0	1
	6	1	1	0
	7	1	1	1
t_2 →	8	0	0	0
	9	0	0	1
	...	...	...	...

Table 6.35

After 8th clock pulses states the pattern is repeated.

MOD of the counter = $t_2 - t_1 = 8 - 0 = 8$

So, this is a MOD-8 up counter.

Note:

- Negative (–ve) edge trigger and Q as a clock → Up counter.
- Negative (–ve) edge trigger and $\overline{Q}$ as a clock → Down counter.
- Positive (+ve) edge trigger and Q as a clock → Down counter.
- Positive (+ve) edge trigger and $\overline{Q}$ as a clock → Up counter.

Above 3-bit Ripple up counter has 8 used state, so it is MOD-8 counter.

Previous Years' Questions

A 4-bit modulo-16 ripple counter uses J-K flip-flops. If the propagation delay of each FF is 50 ns, the maximum clock frequency that can be used is equal to:

(A) 20 MHz (B) 10 MHz

(C) 5 MHz (D) 4 MHz

Solution: (C)

CLK	Q_2	Q_1	Q_0
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0 ($2t_{pd}$)	1 (t_{pd})
6	1 ($3t_{pd}$)	1	0
7	1	1	1
8	0	0	0

Table 6.36

3-Bit Ripple Up Counter (Positive Edge-Triggered Clock)

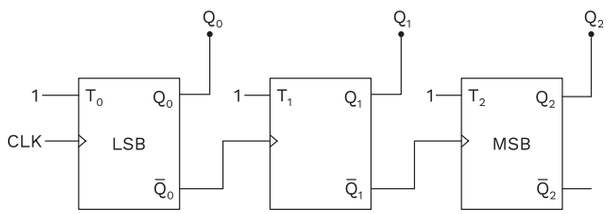


Fig. 6.81

If every flip-flop has a propagation delay of tpd. then for 3-bit Ripple counter total delay = $3t_{pd}$. For n-bit Ripple counter total delay = nt_{pd}

State diagram

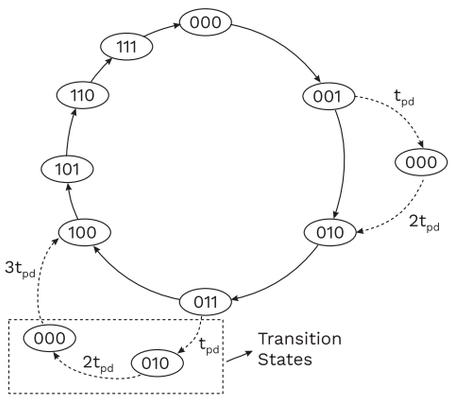


Fig. 6.82

In the state diagram of 3-bit Ripple up counter between the state 011 to 100, there will be two transition states, after t_{pd} delay LSB bit of 011 changes to 010, after $2t_{pd}$ delay output Q_1 will toggle so, 010 changes to 000, at $3t_{pd}$ delay output Q_2 will toggle, so the output changes from, 000 to 100.

So there will be two transition states (010 and 000) between 011 to 100, which takes $3t_{pd}$ delay to produce the next state.

These invalid states are called transition states and the error occurs due to these states are called as decoding errors.

Previous Years' Questions

A 0 to 6 counter consists of 3 flip-flops and a combinational circuit of 2 input gate(s). The combination circuit consists of:

(A) One AND gate

(B) One OR gate

(C) One AND gate and OR gate

(D) Two AND gates

Solution: (D)

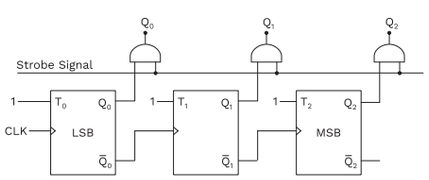


Fig. 6.83

Delay	States (011 to 100) Before Strobe Signal			States (011 to 100) After Strobe Signal		
	Q_2	Q_1	Q_0	Q_2	Q_1	Q_0
	0	1	1	0	1	1
t_{pd}	0	1	0	0	0	0
$2t_{pd}$	0	0	0	0	0	0
$3t_{pd}$	1	0	0	1	0	0

There will be a transition state but it does not reflect on output.

Table 6.37 3- Bit Ripple UP Counter (Positive Edge-triggered Clock)

- The disadvantage of the ripple counter is that transition states are present due to delay of flip-flop (Decoding errors).
- If only one flip-flop changes its state, then no transition states will be present, if more than one flip-flop changes its states, then only transition states present.
- To avoid decoding errors strobe signal is used.
- The strobe signal is kept 'low' for $3t_{pd}$, for 3-bit counter, so that transition states are not reflected and after $3t_{pd}$ time strobe signal becomes 'high'.

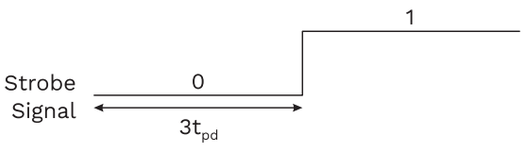


Fig. 6.84

- For 3-bit Ripple counter (Asynchronous Counter), we hold the strobe signal to logic '0' (low) for $3t_{pd}$ time, if there are n-bit ripple counter (Asynchronous counter) we have to hold the strobe signal to logic '0' (low) for ntpd time, which is not feasible that is why we use synchronous counter instead of the asynchronous counter.

3-Bit Asynchronous Down Counter/3-Bit Ripple Counter

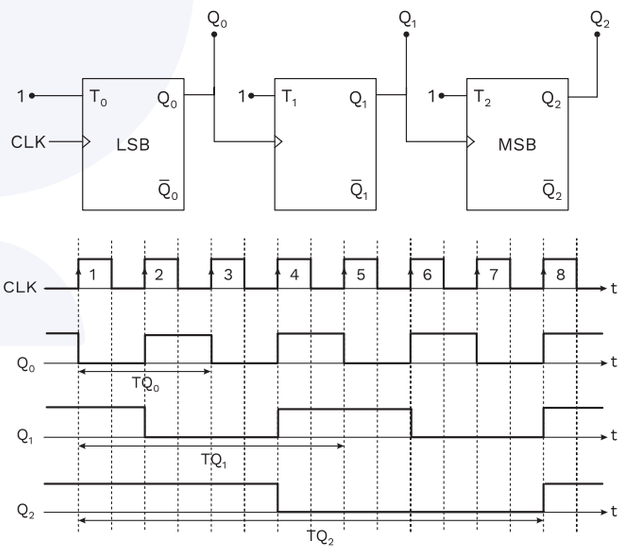


Fig. 6.85

So, $TQ_0 = 2T_{CLK}$

$TQ_1 = 4T_{CLK}$

$TQ_2 = 8T_{CLK}$

- Output Q_0 will change (toggles) for every positive edge of clock.
- Output Q_1 will change (toggles) for every positive edge of output Q_0 (Q_0 changes $0 \rightarrow 1$, Q_1 toggles)

- Output Q_2 will change (toggles) for every positive edge of output Q_1 (Q_1 changes 0→ 1, Q_2 toggles).

	CLK	Q_2	Q_1	Q_0
t_1→	0	1	1	1
	1	1	1	0
	2	1	0	1
	3	1	0	0
	4	0	1	1
	5	0	1	0
	6	0	0	1
	7	0	0	0
t_2→	8	1	1	1
	9	.....	.....	.....

Table 6.38

MOD of the counter = $t_2 - t_1 = 8 - 0 = 8$
So, this is MOD-8 down counter.

3-Bit Ripple Up-Down Counter

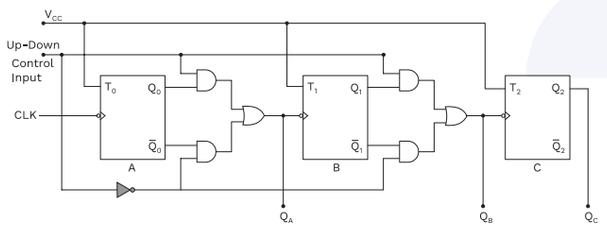


Fig. 6.86

Ripple Up-down counter is controlled by the control input.
If, Control = 0 → Down Counter
Control = 1 → Up Counter

Previous Years' Questions

A 4-bit ripple counter and a 4-bit synchronous counter is made using flip flops having a propagation delay of 10 ns each. If the worst-case delay in the ripple counter and the synchronous counter be R and S respectively, then

(A) R = 10ns, S = 40 ns
(B) R = 40ns, S = 10 ns
(C) R = 10ns, S = 30 ns
(D) R = 30ns, S = 10 ns
Solution: (B)

Down Counter			
Clock	Output		
	Q_C	Q_B	Q_A
t_1→ 0	1	1	1
1	1	1	0
2	1	0	1
3	1	0	0
4	0	1	1
5	0	1	0
6	0	0	1
7	0	0	0
t_2→ 8	1	1	1

Table 6.39

Down Counter			
Clock	Output		
	Q_C	Q_B	Q_A
0	1	1	1
t_1→ 1	1	1	0
2	1	0	1
3	1	0	0
4	0	1	1
5	0	1	0
6	0	0	1
7	0	0	0
t_2→ 8	1	1	1

Table 6.40

Previous Years' Questions

Two D flip-flops, as shown below, are to be connected as a synchronous counter that goes through the following Q1Q0 sequence 00 → 01 → 11 → 10 → 00 → ...

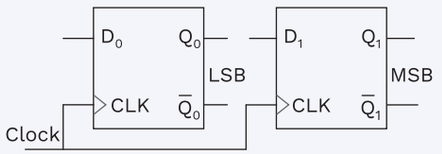


Fig. 6.86

The inputs D0 and D1 respectively should be connected as:

(A) $\bar{Q}_1$ and Q_0

(B) $\bar{Q}_0$ and Q_0

(C) $\bar{Q}_1Q_0$ and $\bar{Q}_1Q_0$

(D) $\bar{Q}_1\bar{Q}_0$ and Q_1Q_0

Solution: (A)

Timing diagram:

1. Up counter

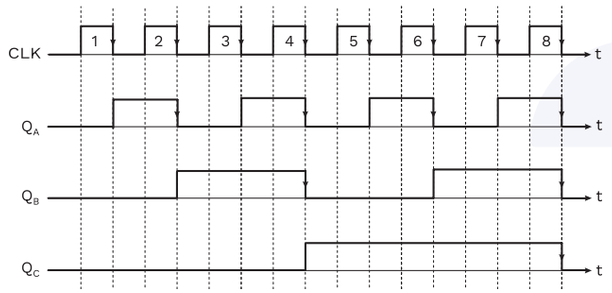


Fig. 6.87

2. Down counter

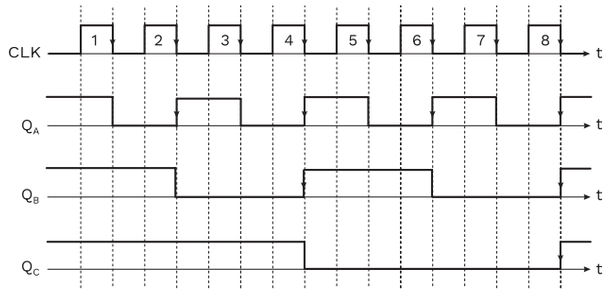


Fig. 6.88

Binary Counter

- If the counter counts all the possible state (Up or Down), then it is called 'Binary counter'.
- In binary counter, there will be no invalid state.
- 3-Bit Binary counter states.

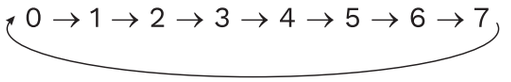


Fig. 6.89

Non-Binary Counter

- If the counter counts only desired states (Up and Down), then these type of counter is called 'Non-Binary counter'.
- In Non-Binary counter, there will be one or more invalid states present.
- 3-bit Non-Binary counter states,

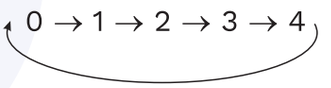


Fig. 6.90

So, 5, 6 and 7 are invalid states.

Asynchronous Clear and Asynchronous Preset

CLEAR control is used to reset the flip-flop.

PRESET control is used to set the flip-flop.

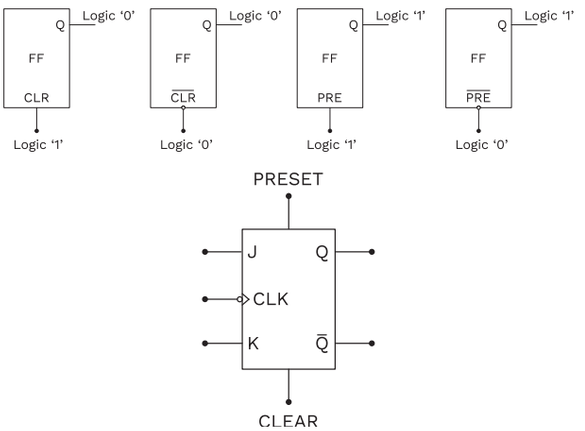


Fig. 6.91

Preset	Clear	Flip-Flop Response
0	0	Clocked Operation
0	1	0 (Reset)
1	0	1 (Set)
1	1	Invalid

Table 6.41 Asynchronous Clear and Asynchronous PRESET

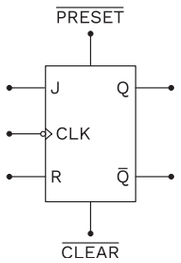


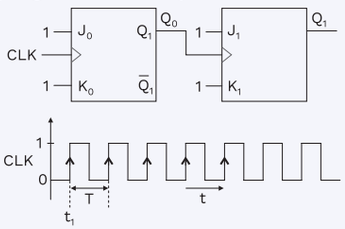
Fig. 6.92

Preset	Clear	Flip-Flop Response
0	0	Invalid
0	1	1 (Set)
1	0	0 (Reset)
1	1	Clock Operation

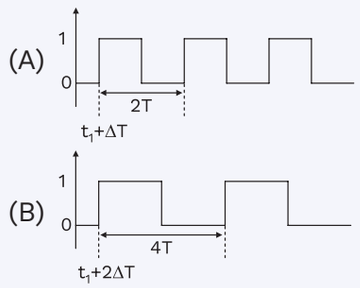
Table 6.42 Asynchronous Clear(Bar) and Asynchronous PRESET(Bar)

Previous Years' Questions

For each of the positive edge-triggered JK flip flop used in the following figure, the propagation delay is ΔT.



Which of the following waveforms correctly represents the output at Q_1?



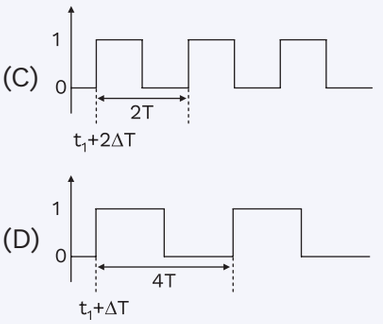


Fig. 6.93

Solution: (B)

Example 20:

Design a MOD-6 asynchronous up-counter using AND gate.

Solution:

MOD-6 counter requires 3 flip-flops.

Let flip-flops outputs are Q_2, Q_1 and Q_0.

CLK	Q_2	Q_1	Q_0	Reset (R)
0	0	0	0	0
1	0	0	1	0
2	0	1	0	0
3	0	1	1	0
4	1	0	0	0
5	1	0	1	0
6	1 ↓ reset 0	1 ↓ reset 0	0 ↓ reset 0	1

Table 6.43

Previous Years' Questions

A switch-tail ring counter is made by using a single D flip-flop. The resulting circuit is a:

(A) S-R flip-flop
(B) J-K flip-flop
(C) D flip-flop
(D) T flip-flop

Solution: (C)

At 6th clock pulse reset control is activated and the output of the counter is reset to 000.

Counter will reset, $R = Q_2\, Q_1\, \overline{Q}_0$

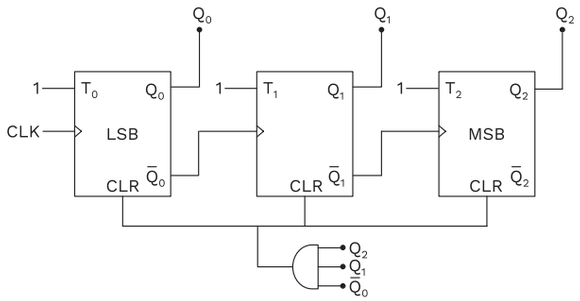


Fig. 6.94

Example 21:

Design a MOD-5 asynchronous up-counter using AND and NAND gate.

Previous Years' Questions

A ripple counter using negative edge triggered D-flip flops is shown in the figure below. The flip-flops are cleared to '0' at the R input. The feedback logic is to be designed to obtain the count sequence shown in the same figure:

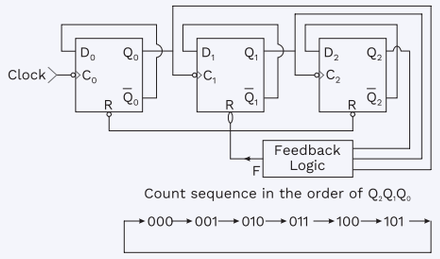


Fig. 6.95

The correct feedback logic is:

(A) $F = \overline{Q_2 Q_1 \overline{Q}_0}$

(B) $F = Q_2 \overline{Q}_1 \overline{Q}_0$

(C) $F = \overline{Q}_2 \overline{Q}_1 Q_0$

(D) $F = \overline{Q}_2 \overline{Q}_1 \overline{Q}_0$

Solution: (A)

Solution:

MOD-5 counter requires 3 flip-flops.

Let flip-flops outputs are Q_2, Q_1 and Q_0.

CLK	Q_2	Q_1	Q_0	Reset (R)
0	0	0	0	0
1	0	0	1	0
2	0	1	0	0
3	0	1	1	0
4	1	0	0	0
5	1 ↓ reset 0	0 ↓ reset 0	1 ↓ reset 0	1

Table 6.44

Previous Years' Questions

Given below figure shows a MOD-k counter, here k is equal to

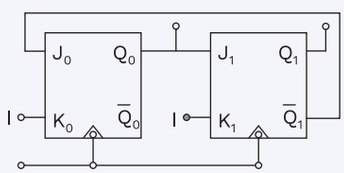


Fig. 6.96

(A) 1 (B) 2

(C) 3 (D) 4

Solution: (C)

At the 5th clock pulse reset control is activated and the output of the counter is reset to 000.

Counter design using AND gate,

Counter will reset, $R = Q_2\, \overline{Q}_1\, Q_0$

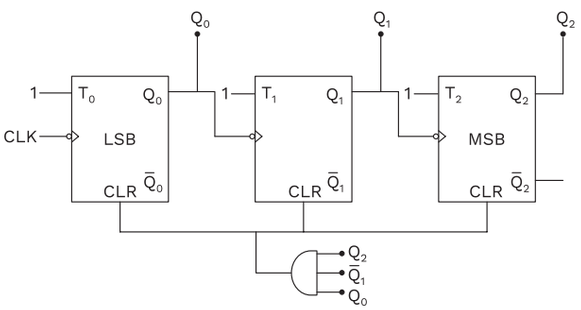


Fig. 6.97

Counter design using NAND gate,

$R = Q_2 \bar{Q}_1 Q_0$

$\bar{R} = \overline{Q_2 \bar{Q}_1 Q_0}$

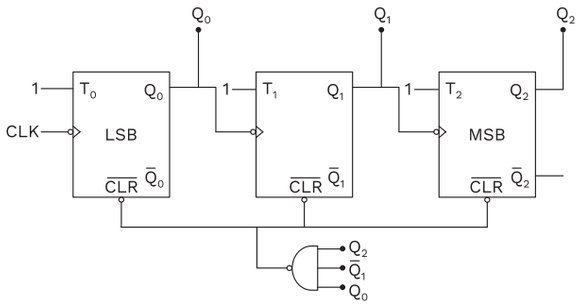


Fig. 6.98

Previous Years' Questions

What is the counting state (Q_1, Q_2) for the counter shown in the figure below?

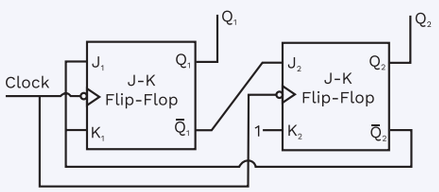


Fig. 6.99

(A) 11, 10, 00, 11, 10, ...
(B) 01, 10, 11, 00, 01, ...
(C) 00, 11,01, 10, 00, ...
(D) 01, 10, 00, 01, 10, ...

Solution: (A)

Delay Analysis Of Asynchronous Counter:

For n-bits asynchronous counter, n-flip-flops are required.

So, Delay = nt_{pd}

TCLK ≥ Delay

TCLK ≥ nt_{pd}

$\frac{1}{f_{CLK}} \geq nt_{pd}$

$f_{CLK} \leq \frac{1}{nt_{pd}}$

Example 22:

An asynchronous counter is given below, find:

1. MOD of the counter.
2. Output frequency of the counter, if the input frequency is 10 MHz.
3. If the delay of each flip-flop is 50 nsec, then the maximum frequency of the clock.

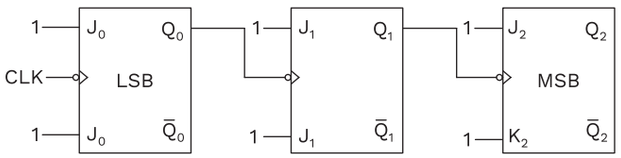


Fig. 6.99

Solution:

1. All the J-K flip-flop is in toggle mode.
   MOD number = $2^n = 2^3$
   MOD number = 8
2. Output frequency of counter,
   fin = 10 MHz

$$f_{Q_0} = \frac{10}{2} = 5\,MHz$$

$$f_{Q_1} = \frac{5}{2} = 2.5\,MHz$$

$$f_{Q_0} = \frac{2.5}{2} = 1.25\,MHz$$

3. Delay = $3t_{pd}$
   Delay = 3(50) = 150 nsec

Maximum frequency, $f_{CLK} \leq \frac{1}{150 \times 10^{-9}}$

f_{CLK} = 6.66 MHz

Previous Years' Questions

For the circuit shown, the counter state (Q_1Q_0) follows the sequence:

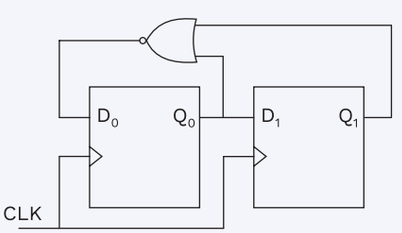


Fig. 6.99

(A) 00, 01, 10, 11, 00
(B) 00, 01, 10, 00, 01
(C) 00, 01, 11, 00, 01
(D) 00, 10, 11, 00, 10

Solution: (B)

Example 23:

Design an asynchronous BCD up counter using AND gate.

Solution:

BCD up counter counts from 0(0000) to 9(1001).

So, we required 4-flip-flops in toggle mode and there will be 10 states.

CLK	Q_3	Q_2	Q_1	Q_0	State Numbers
Initially	0	0	0	0	–
1	0	0	0	1	1
2	0	0	1	0	2
3	0	0	1	1	3
4	0	1	0	0	4
5	0	1	0	1	5
6	0	1	1	0	6
7	0	1	1	1	7
8	1	0	0	0	8
9	1	0	0	1	9
10	0	0	0	0	0

Table 6.45

Reset (R) = $Q_3\bar{Q}_2Q_1\bar{Q}_0$

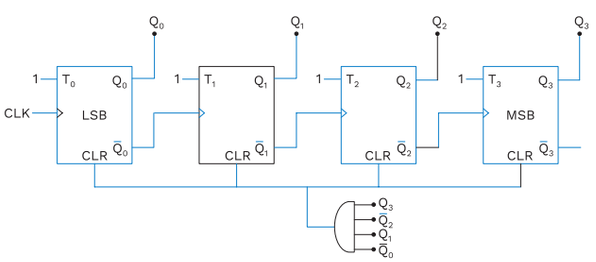


Fig. 6.100

Synchronous Counter

In sequential circuits, there are three types synchronous counter:

1. Ring counter
2. Johnson ring counter
3. User defined counter

4-bit ring counter:

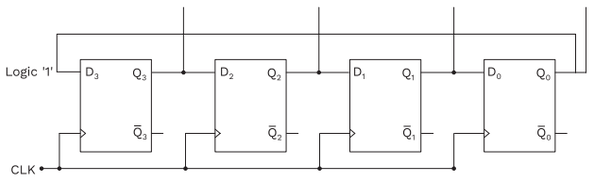


Fig. 6.101

- Ring counter is a synchronous counter, it is a shift register in which last flip-flop output is connected to the first flip-flop input.
- In ring counter, only one flip-flop is logic '1' and it will rotate with a clock.
- Ring counter performs right shift operation.

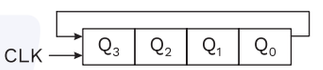


Fig. 6.102

- Let the initial output of the ring counter $Q_3Q_2Q_1Q_0$ be 1000.

CLK	Q_3	Q_2	Q_1	Q_0	Decoding Logic
$t_1 \rightarrow 0$	1	0	0	0	Q_3
1	0	1	0	0	Q_2
2	0	0	1	0	Q_1
3	0	0	0	1	Q_0
$t_2 \rightarrow 4$	1	0	0	0	Q_3

Table 6.46 Truth Table of Synchronous Counter

MOD number of counter = $t_2 - t_1 = 4$
Hence, it is a MOD-4 counter.

State Diagram:

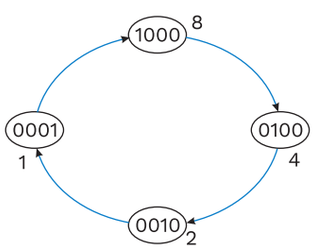


Fig. 6.103

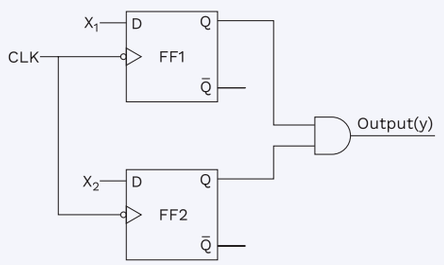
- Valid states of 4-bit Ring counter = 1, 2, 4, 8
- Invalid states of 4-bit Ring counter = 0, 3, 5, 6, 7, 9, 10, 11, 12, 13, 14, 15.
- Repeating pattern, for output Q3,

CLK	Q_3
0	1
1	0
2	0
3	0
4	1

Table 6.47 State Diagram

Previous Years' Questions

In the circuit shown, choose the correct timing diagram of the output (y) from the given waveforms W_1, W_2, W_3 and W_4:



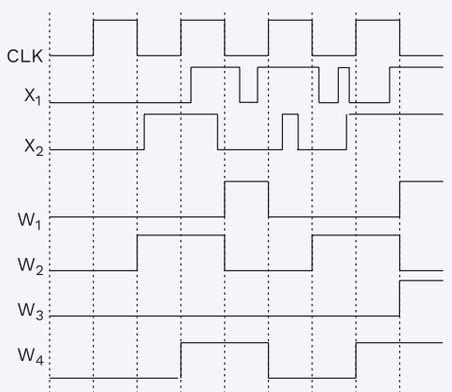


Fig. 6.104

(A) W_1 (B) W_2
(C) W_3 (D) W_4

Solution: (C)

$T_{Q_3} = 4T_{CLK}$

$f_{Q_3} = \frac{f_{CLK}}{4}$

Duty Cycle,

$D = \frac{T_{ON}}{T_{ON} + T_{OFF}} \times 100$

$D = \frac{1}{4} \times 100$

D = 25%

- For Output Q2,

CLK	Q_2
0	0
1	1
2	0
3	0
4	0

Table 6.48

Previous Years' Questions

The figure shows a binary counter with synchronous clear input. With the decoding logic shown, the counter works as a:

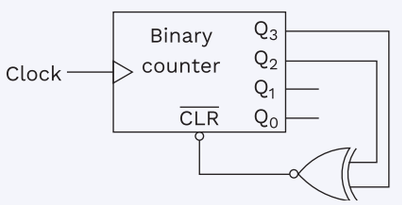


Fig. 6.105

(A) MOD-2 counter
(B) MOD-4 counter
(C) MOD-5 counter
(D) MOD-6 counter

Solution: (C)

$T_{Q_2} = 4T_{CLK}$

$f_{Q_2} = \frac{f_{CLK}}{4}$

Duty cycle,

$$D = \frac{T_{ON}}{T_{ON} + T_{OFF}} \times 100$$

D = 25%

- For Output Q_1,

CLK	Q_1
0	0
1	0
2	1
3	0
4	0

Table 6.49

$T_{Q_1} = 4T_{CLK}$

$f_{Q_1} = \frac{f_{CLK}}{4}$

Duty cycle, (D) = 25%

Previous Years' Questions

Assume that all the digital gates in the circuits shown in the figure are ideal, the resistor $R = 10k\Omega$ and the supply voltage is 5V. The D flip-flops D_1, D_2, D_3, D_4 and D_5 are initialized with logic values 0, 1, 0, 1 and 0 respectively. The clock has a 30% duty cycle:

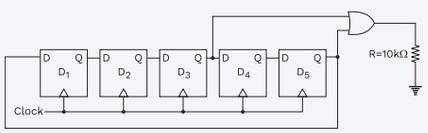


Fig. 6.106

The average power dissipated (in mW) in the resistor R is_____.

Solution: Range (1.45-1.55)

- For Output Q0,

$T_{Q_0} = 4T_{CLK}$

$f_{Q_0} = \frac{f_{CLK}}{4}$

Duty cycle, (D) = 25%

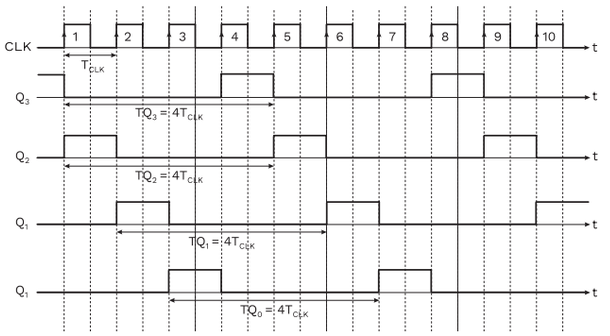


Fig. 6.107

For n-bit Ring counter,

- Number of used states = n
- Number of unused states = 2n – n
- The phase shift between successive waveform $= \frac{360°}{n}$
- If the delay of each flip-flop is tpd, then $T_{CLK} \geq (t_{pd})_{FF}$
- If the flip-flops have a different delay then $T_{CLK} \geq (t_{pd})_{max}$

Previous Years' Questions

Two D flip-flops are connected as a synchronous counter that goes through the following Q_BQ_A sequence

$00 \rightarrow 11 \rightarrow 01 \rightarrow 10 \rightarrow 00 \rightarrow \ldots$

The connections to the inputs D_A and D_B are:

(A) $D_A = Q_B, D_B = Q_A$

(B) $D_A = \overline{Q}_A, D_B = \overline{Q}_B$

(C) $D_A = \left(Q_A\overline{Q}_B + \overline{Q}_AQ_B\right), D_B = Q_A$

(D) $D_A = \left(Q_AQ_B + \overline{Q}_A\overline{Q}_B\right), D_B = \overline{Q}_B$

Solution: (D)

Advantage Of Ring Counter:

Decoding logic of the ring counter is simple and does not require any external logic circuit.

Drawbacks Of Ring Counter:

- If all the outputs of flip-flops are initially zero, then the Ring counter does not start.

- If the 2 flip-flop outputs' are high initially, then the ring counter enters into an unused state and never come out of an unused state; this is called as **lockout problem.**

Johnson Ring Counter:

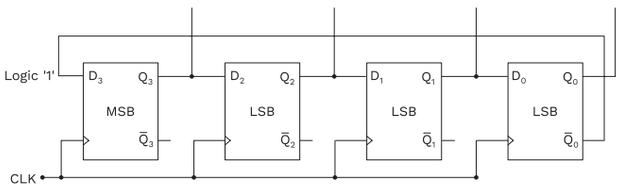
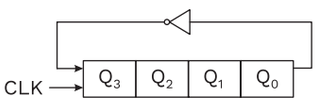


Fig. 6.108

- Johnson counter also works as shift right register.



- Let the initial output Q_3 Q_2 Q_1 Q_0 of Johnson counter be 0000.

CLK	Q_3 Q_2 Q_1 Q_0	Decoding Logic
0	0 0 0 0	$\bar{Q}_3$ $\bar{Q}_2$ $\bar{Q}_1$ $\bar{Q}_0$
1	1 0 0 0	Q_3 $\bar{Q}_2$ $\bar{Q}_1$ $\bar{Q}_0$
2	1 1 0 0	Q_3 Q_2 $\bar{Q}_1$ $\bar{Q}_0$
3	1 1 1 0	Q_3 Q_2 Q_1 $\bar{Q}_0$
4	1 1 1 1	Q_3 Q_2 Q_1 Q_0
5	0 1 1 1	$\bar{Q}_3$ Q_2 Q_1 Q_0
6	0 0 1 1	$\bar{Q}_3$ $\bar{Q}_2$ Q_1 Q_0
7	0 0 0 1	$\bar{Q}_3$ $\bar{Q}_2$ $\bar{Q}_1$ Q_0
8	0 0 0 0	$\bar{Q}_3$ $\bar{Q}_2$ $\bar{Q}_1$ $\bar{Q}_0$

Table 6.50

Used states of Johnson counter

= 0, 1, 3, 8, 12, 14, 7, 15

Unused states of Johnson counter

= 2, 4, 5, 6, 9, 10, 11, 13

Repeating Pattern Of Johnson Ring Counter:

Q_3	Q_2	Q_1	Q_0
$TQ_3 = 8T_{CLK}$	$TQ_2 = 8T_{CLK}$	$TQ_1 = 8T_{CLK}$	$TQ_0 = 8T_{CLK}$
$fQ_3 = \frac{f_{CLK}}{8}$	$fQ_2 = \frac{f_{CLK}}{8}$	$fQ_1 = \frac{f_{CLK}}{8}$	$fQ_0 = \frac{f_{CLK}}{8}$
$D = \frac{1}{2} \times 100\% = 50\%$	$D = \frac{1}{2} \times 100\% = 50\%$	$D = \frac{1}{2} \times 100\% = 50\%$	$D = \frac{1}{2} \times 100\% = 50\%$

Table 6.51

For 4-Bit Johnson Counters:

- Numbers of used states = 8
- Numbers of unused state = 8
- MOD of counter = 8
- Frequency of each flip-flop = $\frac{f_{CLK}}{8}$

Previous Years' Questions

For the circuit shown in the figure, the delay of the bubbled NAND gate is 2 ns and that of the counter is assumed to be zero.

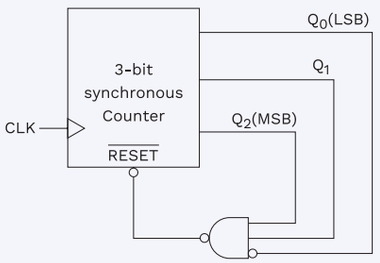


Fig. 6.109

If the clock (CLK) frequency is 1 GHz, then the counter behaves as a:

(A) MOD-5 counter
(B) MOD-6 counter
(C) MOD-7 counter
(D) MOD-8 counter

Solution: (D)

For n-bit Johnson counters:

- Number of used states = 2n
- Numbers of unused states = 2n – 2n
- MOD of counter = 2n
- Frequency of each flip-flop = $\frac{f_{CLK}}{2n}$

Example 24:

What happens when Johnson counter enters one of its unused states.

Solution:

Used states of Johnson counter

= 0, 1, 3, 8, 12, 14, 7, 15

Unused states of Johnson counter

= 2, 4, 5, 6, 9, 10, 11, 13

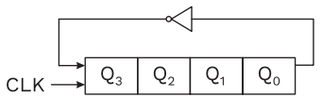


Fig. 6.110

Let's assume Johnson counter enters 2nd state.

CLK	Q_3	Q_2	Q_1	Q_0	States
0	0	0	1	0	2
1	1	0	0	1	9
2	0	1	0	0	4
3	1	0	1	0	10
4	1	1	0	1	13
5	0	1	1	0	6
6	1	0	1	1	11

Table 6.52

If the Johnson ring counter enters the unused states, then lockout problem occurs.

- **Drawback of Johnson ring counter:**
  If the Johnson counter enters any of its unused state, then counter permanently stayed in the unused states only.

Previous Years' Questions

A MOD-n counter using a synchronous binary up-counter with synchronous clear input is shown in the figure. The value of n is_____.

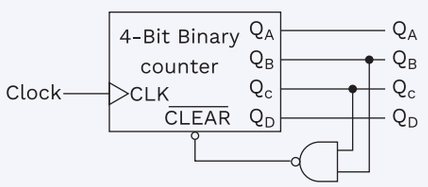


Fig. 6.111

Solution: 7

Modified Johnson ring counter to prevent lockout problem:

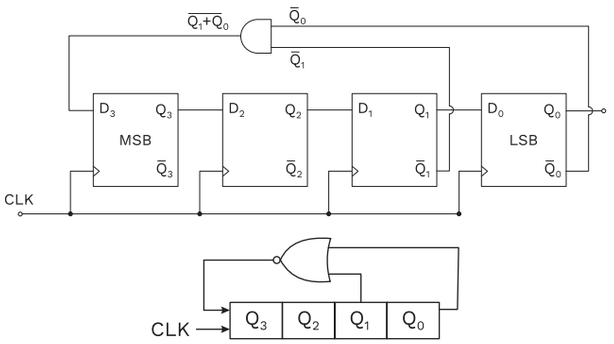


Fig. 6.112

Let's assume Johnson counter enters 9th state.

CLK	Q_3	Q_2	Q_1	Q_0	States
0	1	0	0	1	9
1	0	1	0	0	4
2	1	0	1	0	10
3	0	1	0	1	5
4	0	0	1	0	2
5	0	0	0	1	1
6	0	0	0	0	0
7	1	0	0	0	8
8	1	1	0	0	12
9	1	1	1	0	14
10	0	1	1	1	7
11	0	0	1	1	3
12	0	0	0	1	1
13	0	0	0	0	0
14	1	0	0	0	8
15	1	1	0	0	12

Table 6.53 State Diagram of Johnson Counter

Previous Years' Questions

The digital logic shown in the figure satisfies the given state diagram when Q_1 is connected to input A of the XOR gate.

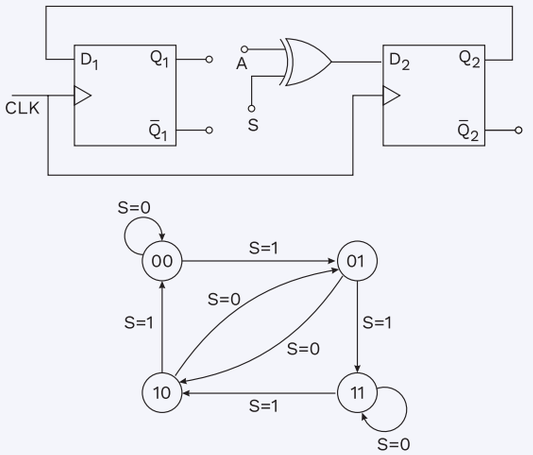


Fig. 6.113

Suppose the XOR gate is replaced by an XNOR gate. Which one of the following option preserves the state diagram?

(A) Input A is connected to $\overline{Q}_2$.

(B) Input A is connected to Q_2.

(C) Input A is connected to $\overline{Q}_1$ and S is complemented.

(D) Input A is connected to $\overline{Q}_1$.

Solution: (D)

In modified Johnson ring counter, if counter enters into the unused state. It comes out of an unused state.

Ring Counter	Johnson Ring Counter
MOD number = n	MOD number = 2n
Number of used states = n Number of unused states = 2^n-n	Number of used states = 2n Number of unused states = 2^n-2n
Time period of each flip-flop = nT_{CLK}	Time period of each flip-flop = $(2n)T_{CLK}$

Ring Counter	Johnson Ring Counter
Frequency of each flip-flop = $\frac{f_{CLK}}{n}$	Frequency of each flip-flop = $\frac{f_{CLK}}{2n}$
Suffer from lockout problem	Suffer from lockout problem
Decoding logic is simple	Decoding logic required AND and NOR gates

Table 6.54

Previous Years' Questions

The propagation delay of the exclusive OR (XOR) gate in the circuit is 3 ns. The propagation delay of all the flip-flops is assumed to be zero. The clock (CLK) frequency provided to the circuit is 500 MHz.

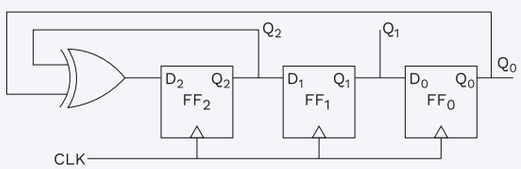


Fig. 6.114

Starting from the initial value of flip-flop outputs $Q_2Q_1Q_0 = 111$ with $D_2 = 1$. The minimum number of triggering clock edges after which the flip-clop outputs $Q_2Q_1Q_0$ becomes 100 (in integer) is _____.

Solution: 5

1. Johnson ring counter has different names:
   a) Johnson ring counter
   b) Twisted ring counter
   c) Switch tail counter
   d) Walking counter
   e) Creeping counter
   f) Mobies counter

Example 25:

Design synchronous counter, whose counting sequence is Q1Q0 = 00 → 10 → 01 → 11 → 00 → 10 .... by using T-flip-flop.

Solution:

Given, 00 → 10 → 01 → 11 → 00 → 10 ...

used states,

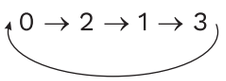


Fig. 6.115

Mod of counter = 4 = 2n,

Number of flip-flops required, n = 2

So we require two T-flip-flops.

Previous Years' Questions

When the output Y in the circuit below is "1", which implies that data has:

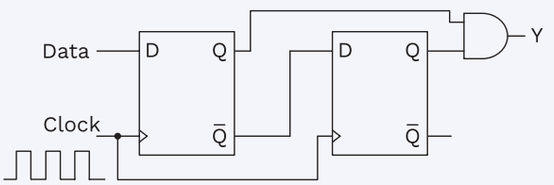


Fig. 6.116

(A) Changed from "0" to "1"
(B) Changed from "1" to "0"
(C) Changed in either direction
(D) Not changed

Solution: (A)

Present States		Next States		Flip-Flops Inputs	
Q_1	Q_0	Q_1^+	Q_0^+	T_1	T_0
0	0	1	0	1	0
1	0	0	1	1	1
0	1	1	1	1	0
1	1	0	0	1	1

Input value depends upon Q_1 and Q_1^+ by excitation table.

Input value depends upon Q_0 and Q_0^+ by excitation table.

Table 6.55

T1 = 1

and $T_0 = Q_1\bar{Q}_0 + Q_1Q_0$

T0 = Q1

Implementation of counter

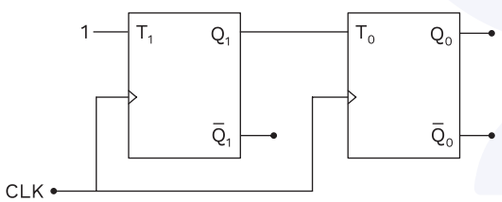


Fig. 6.117

Example 26:

Design synchronous counter, whose counting sequence is

Q1Q0 = 00 → 10 → 01 → 11 → 00 → 10 .... 00 By using D-flip-flops

Solution:

Given, Q1Q0 = 00 → 10 → 01 → 11 → 00 → 10 00

Used states,

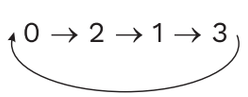


Fig. 6.118

Previous Years' Questions

A 4-bit shift register circuit configured for right-shift operation,
i.e., $D_{in} \rightarrow A, A \rightarrow B, B \rightarrow C, C \rightarrow D,$ is shown.

If the present state of the shift register is ABCD = 1101, the number of clock cycles required to reach the state ABCD = 1111 is _____.

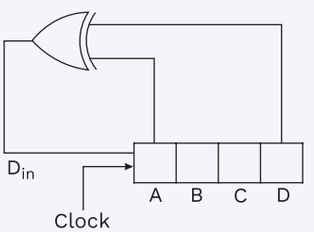


Fig. 6.119

Solution: 10

Present States		Next States		Flip-Flops Inputs	
Q_1	Q_0	Q_1^+	Q_0^+	D_1	D_0
0	0	1	0	1	0
1	0	0	1	0	1

Present States		Next States		Flip-Flops Inputs	
Q_1	Q_0	Q_1^+	Q_0^+	D_1	D_0
0	1	1	1	1	1
1	1	0	0	0	0

Table 6.56

$D_1 = \bar{Q}_1\bar{Q}_0 + \bar{Q}_1Q_0$

$D_1 = \bar{Q}_1$

D0 = Q1 Å Q0

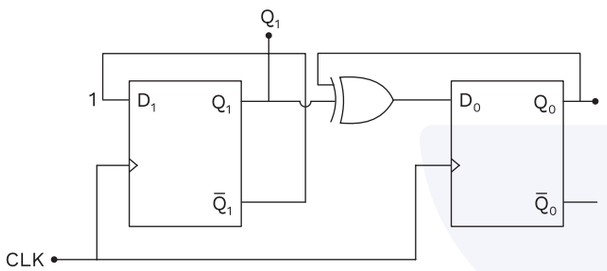


Fig. 6.120

Previous Years' Questions

A three-bit pseudo-random number generator is shown. Initially the value of the output $Y = Y_2Y_1Y_0$ is set to 111. The value of output Y after three clock cycles are:

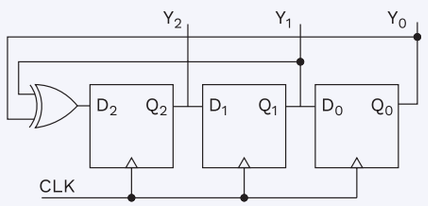


Fig. 6.121

(A) 000 (B) 001

(C) 010 (D) 100

Solution: (D)

Setup time: It is the minimum amount of time before the active edge of the clock; the input signal should remain constant.

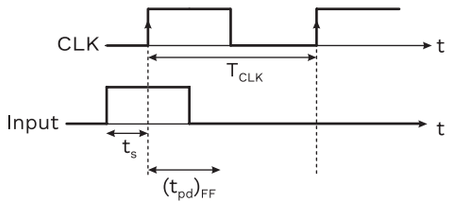


Fig. 6.122

Hold time: It is the minimum amount of time after the active edge of the clock, the input signal should remain constant.

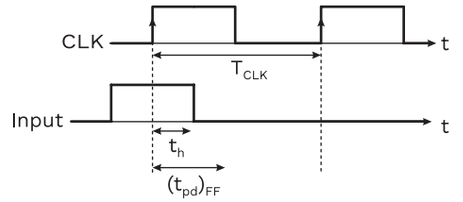


Fig. 6.123

Delay in single flip-flop

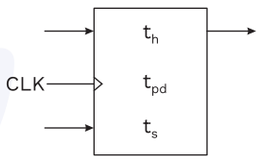


Fig. 6.124

$Delay = t_{pd} + t_s$

$T_{CLK} \geq t_{pd} + t_s$

Delay in synchronous counter without feedback:

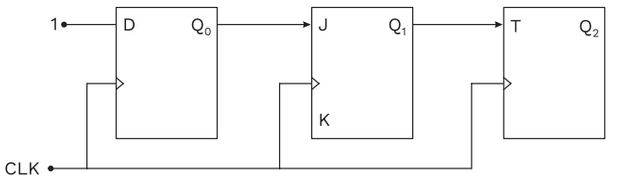


Fig. 6.125

$T_{CLK} \geq (t_{pd})_{FF} + t_s$

$T_{CLK} \geq (t_{pd})_{max} + (t_s)_{max}$

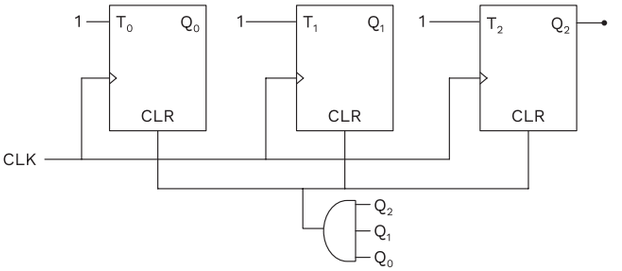


Fig. 6.126

$T_{CLK} \geq (t_{pd})_{FF} + t_s + (t_{pd})_{AND}$

Previous Years' Questions

In the circuit shown below, a positive edge-triggered D flip-flop is used for sampling input data D_{in} using clock CK.

The XOR gate outputs 3.3 volts for logic HIGH and 0 volts for logic low levels. The data bit and clock periods are equal and the value of $\Delta T / T_{CK} = 0.15$, where the parameters ΔT and T_{CK} are shown in the figure. Assume that the flip-flop and the XOR gate are ideal.

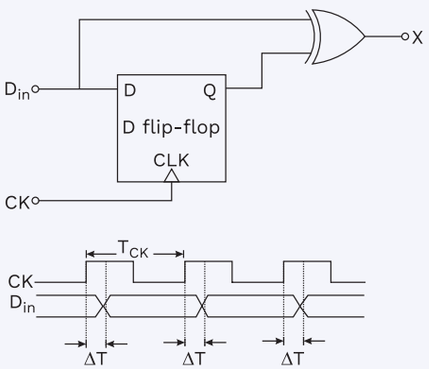


Fig. 6.127

If the probability of input data bit (D_{in}) transition in each clock period is 0.3, the average value (in volts, accurate to two decimal places) of the voltage at node is _____.

Solution: Range (0.82 - 0.86)

Synchronous clear and synchronous PRESET:

- In synchronous counter, clear control used to reset the flip-flop.
- PRESET control is used to set the flip-flop.

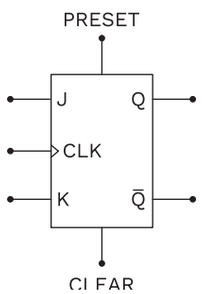


Fig. 6.128

Clear	Preset	CLK	Q^+
0	0	0	Q (Hold)
0	0	1	Clocked Operation
0	1	0	Hold
0	1	1	Set
1	0	0	Hold
1	0	1	Reset
1	1	0	Hold
1	1	1	Invalid

Table 6.57 Truth Table of Synchronous Counter

Previous Years' Questions

The outputs of the two flip-flops, Q_1, Q_2 in the figure shown are initialized to 0, 0. The sequence generated at Q_1 upon application of clock signal is:

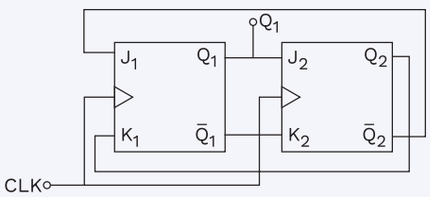


Fig. 6.129

(A) 01110 (B) 01010
(C) 00110 (D) 01100

Solution: (D)

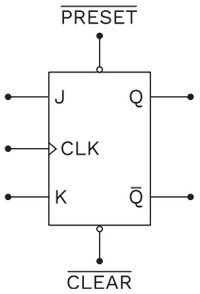
- 

Fig. 6.130

Clear	Present	CLK	Q^+
0	0	0	Hold
0	0	1	Invalid
0	1	0	Hold
0	1	1	Reset
1	0	0	Hold
1	0	1	Set
1	1	0	Hold
1	1	1	Clocked operation

Table 6.58

Example 27:

Draw the output waveform of the JK flip-flop.

1. Asynchronous clear
2. Synchronous clear

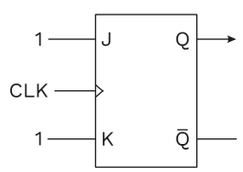


Fig. 6.131

Solution:

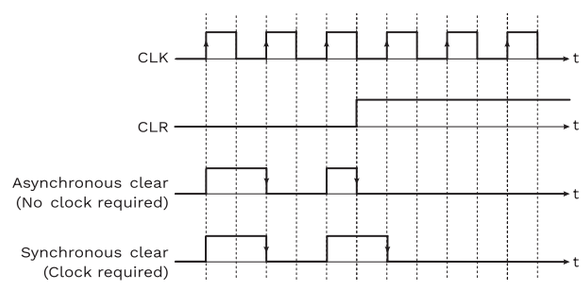


Fig. 6.132

Example 28:

Find the MOD number of counter

1. If $(t_{pd})_{comb} = 0$, Asynchronous clear
2. If $(t_{pd})_{comb} = 0$, Synchronous clear
3. If $(t_{pd})_{comb} = T_{CLK}$, Asynchronous clear
4. If $(t_{pd})_{comb} = T_{CLK}$, Synchronous clear
5. If $(t_{pd})_{comb} < T_{CLK}$, Asynchronous clear
6. If $(t_{pd})_{comb} < T_{CLK}$, Synchronous clear

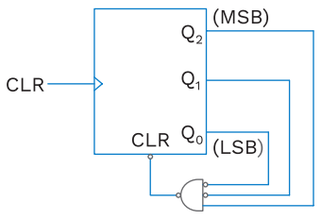


Fig. 6.133

Solution:

Circuit can be redrawn as,

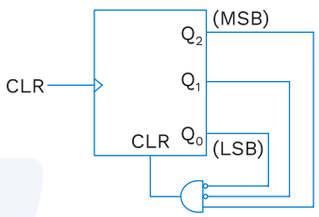


Fig. 6.134

Reset (clear) logic of the circuit,

$$R = Q_2\bar{Q}_1\bar{Q}_0$$

1. $(t_{pd})_{comb} = 0$, Asynchronous clear → No clock required for clear (reset) signal to operate.

CLK	Q_2	Q_1	Q_0	$R = Q_2\bar{Q}_1\bar{Q}_0$
0	0	0	0	0
1	0	0	1	0
2	0	1	0	0
3	0	1	1	0
4	1	0	0	1
	0	0	0	

Immediately clear the output, before 5th clock

Table 6.59

Hence, it is MOD-4 counter.

2. $(t_{pd})_{comb} = 0$, Synchronous clear → Clock required for clear (reset) signal to operate.

CLK	Q_2	Q_1	Q_0	$R = Q_2\bar{Q}_1\bar{Q}_0$
0	0	0	0	0
1	0	0	1	0
2	0	1	0	0
3	0	1	1	0
4	1	0	0	1
	↓			
	0	0	0	

Immediately clear the output ($t_{pd} = 0$), before the 5th clock because clock is present.

Table 6.60

Hence, it is MOD-4 counter.

3. If $(t_{pd})_{comb} = T_{CLK}$, asynchronous clear → No clock required for clear (reset) signal to operate.

CLK	Q_2	Q_1	Q_0	$R = Q_2\bar{Q}_1\bar{Q}_0$
0	0	0	0	0
1	0	0	1	0
2	0	1	0	0
3	0	1	1	0
4	1	0	0	0
5	1	0	1	1
	↓			
	0	0	0	

Clear the output at 5th clock, because $(t_{pd})_{comb} = T_{CLK}$

Table 6.61

Hence, it is a MOD-5 counter.

Previous Years' Questions

A traffic signal cycles from Green to Yellow, Yellow to Red and Red to Green. In each cycle, Green is turned on for 70 seconds. Yellow is turned on for 5 seconds and the Red is turned on for 75 seconds. This traffic light has to be implemented using a finite state machine (FSM). The only input to this FSM is a clock of 5 seconds period. The minimum number flip-flops required to implement this FSM is ____.

Solution: 5

4. If $(t_{pd})_{comb} = T_{CLK}$, Synchronous clear → Clock required for clear (reset) signal to operate.

CLK	Q_2	Q_1	Q_0	$R = Q_2\bar{Q}_1\bar{Q}_0$
0	0	0	0	0
1	0	0	1	0
2	0	1	0	0
3	0	1	1	0
4	1	0	0	0
5	1	0	1	1
	↓			
	0	0	0	

Clear the output at 5th clock, because $(t_{pd})_{comb} = T_{CLK}$

Table 6.62

Hence, it is a MOD-5 counter.

5. If $(t_{pd})_{comb} < T_{CLK}$, Asynchronous clear → No clock required for clear (reset) signal to operate.

Previous Years' Questions

The state transition diagram for a finite state machine with states A, B and C, and binary inputs X, Y and Z is shown in the figure.

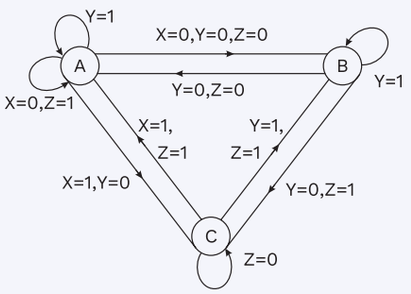


Fig. 6.135

Which one of the following statements is correct?

(A) Transitions from state A are ambiguously defined.

(B) Transitions from state B are ambiguously defined.

(C) Transitions from state C are ambiguously defined.

(D) All of the state transitions are defined unambiguously.

Solution: (C)

CLK	Q_2	Q_1	Q_0	$R = Q_2\bar{Q}_1\bar{Q}_0$
0	0	0	0	0
1	0	0	1	0
2	0	1	0	0
3	0	1	1	0
4	1	0	0	0
$4+t_{pd}$	1	0	1	1
	↓			
	0	0	0	

Clear the output before 5th clock ($4+t_{pd}$), because $(t_{pd})_{comb} < T_{CLK}$

Table 6.63

Hence, it is a MOD-4 counter.

5. If $(t_{pd})_{comb} < T_{CLK}$, Synchronous clear → Clock required for clear (reset) signal to operate.

CLK	Q_2	Q_1	Q_0	$R = Q_2\bar{Q}_1\bar{Q}_0$
0	0	0	0	0
1	0	0	1	0
2	0	1	0	0
3	0	1	1	0
4	1	0	0	0
$4+t_{pd}$	1	0	0	1
5	1	0	1	1
	↓			
	0	0		

Clear the output at 5th clock because synchronous counter required clock for reset operation.

Table 6.64

Hence, it is a MOD-5 counter.

Previous Years' Questions

The state diagram of a finite state machine (FSM) is designed to detect an overlapping sequence of three bits is shown in the figure. The FSM has an input 'In' and output 'Out'. The initial state of the FSM is S_0.

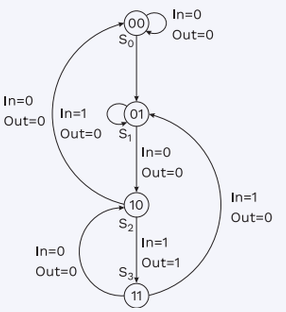


Fig. 6.136

If the input sequence is 10101101001101, starting with the left-most bit, then the number of times 'Out' will be 1 is _____.

Solution: 4

Note:

- If $(t_{pd})_{comb} = nT_{CLK}$.
  MOD number of synchronous counter = MOD number of asynchronous counter.
- If $(t_{pd})_{comb} \neq nT_{CLK}$
  MOD number of synchronous counter = MOD number of asynchronous counter +1.

Some standard assumption for counter problems:

1. Consider up counter, if not mentioned.
2. Q (subscript high) = MSB, if not mentioned.
3. Assume asynchronous CLR, if not mentioned.
4. Assume $(t_{pd})_{comb} < T_{CLK}$, if not mentioned.

Example 29:

A MOD-n counter using a synchronous binary up counter with synchronous clear input is shown in the figure. The value of n is_____.

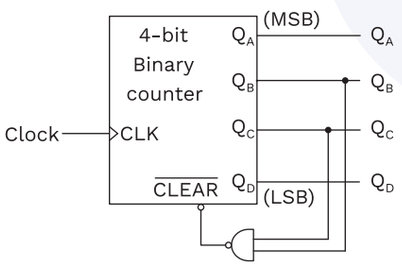


Fig. 6.137

(A) 2 (B) 3 (C) 5 (D) 7

Solution: (D)

Modified circuit:

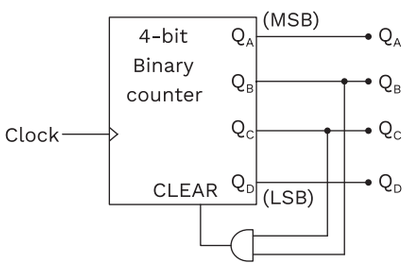


Fig. 6.137

The counter will reset when Q_B and Q_C are logic '1'.

So, $Q_A\ Q_B\ Q_C\ Q_D = 0110$

Considering $t_{pd} < T_{CLK}$, for asynchronous clear MOD = 6.

Considering $t_{pd} < T_{CLK}$, for synchronous clear MOD = 6 + 1 = 7.

Hence, option (D) is correct.

Finite State Machine

1. Synchronous sequential circuit is also called a finite state machine (FSM).
2. There are two types of FSM,
   a) Mealy state machine
   b) Moore state machine

Melay state machine:

The output of Melay state machine is a function of present state as well as present input.

$$Z(t) = f(S(t), x(t))$$

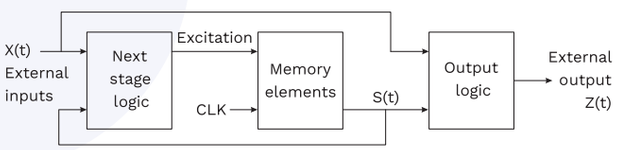


Fig. 6.138

State diagram:

The state diagram or state graph is a pictorial representation of the relationship between the present state, the input, the next state and the output of a sequential circuit, i.e the state diagram is a pictorial representation of the behaviour of a sequential circuit.

State table is given below:

Present State	Next State, Output	
	X = 0	X = 1
a	a, 0	b, 0
b	b, 1	c, 0
c	d, 0	c, 0
d	d, 0	a, 1

Table 6.65 State Table Diagram

State diagram of above state table:

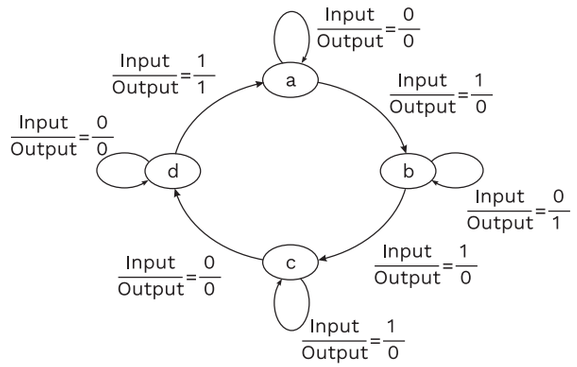


Fig. 6.139

Moore state machine:

The output of the Moore state machine is a function of the present state only.

$$Z(t) = f[s(t)]$$

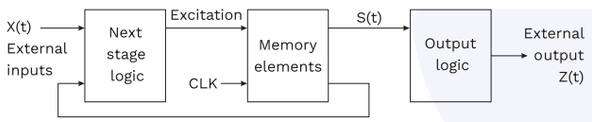


Fig. 6.140

State table given below:

Present State	Next State		Output
	X = 0	X = 1	
a	a	b	0
b	b	c	0
c	d	c	0
d	a	d	1

Table 6.66

State diagram of above state table:

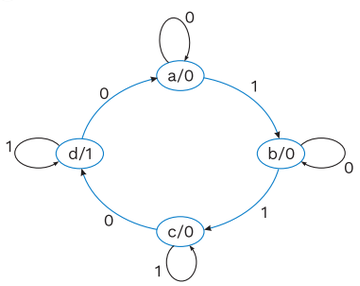


Fig. 6.141

Previous Years' Questions

A finite state machine (FSM) is implemented using the D flip-flops A and B and logic gates, as shown in the figure below. The four possible states of the FSM are $Q_AQ_B = 00, 01, 10$ and 11.

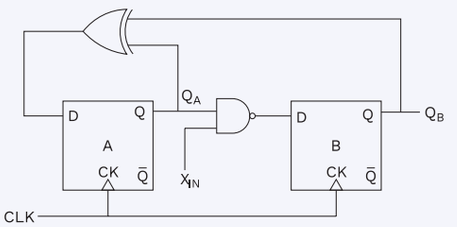


Fig. 6.142

Assume that X_{IN} is held at a constant logic level throughout the operation of FSM. When the FSM is initialized to the state $Q_AQ_B = 00$ and clocked after a few clock cycles, it starts cycling through:

(A) All of the four possible states if $X_{IN} = 1$

(B) Three of the four possible states if $X_{IN} = 0$

(C) Only Two of the four possible state if $X_{IN} = 1$

(D) Only Two of the four possible states if $X_{IN} = 0$

Solution: (D)

Example 30:

Find the reduced state table and reduced state diagram.

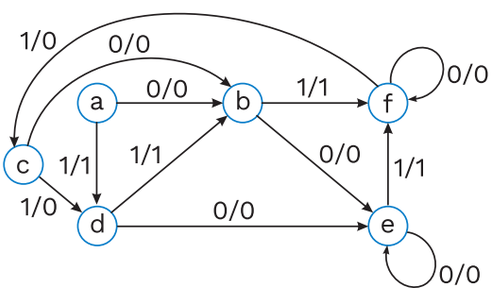


Fig. 6.143

Solution:

State table:

Present State	Next State, Output	
	X = 0	X = 1
a	b, 0	d, 1
b	e, 0	f, 1
c	b, 0	d, 0
d	e, 0	b, 1
e	e, 0	f, 1
f	f, 0	c, 0

Remove

Table 6.67

- Next state and output of state b and e are same, so we can replace e with b from every states.

Reduced state table:

Present State	Next State, Output	
	X = 0	X = 1
a	b, 0	d, 1
b	b, 0	f, 1
c	b, 0	d, 0
d	b, 0	b, 1
f	f, 0	c, 0

Table 6.68

Reduced state diagram

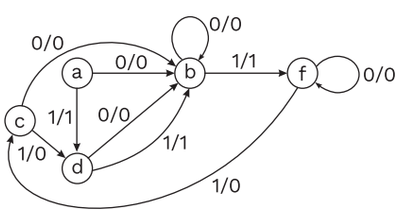


Fig. 6.144

Previous Years' Questions

The state diagram of a sequence detector is shown below. State S_0 is the initial state of the sequence detector. If the output is 1, then

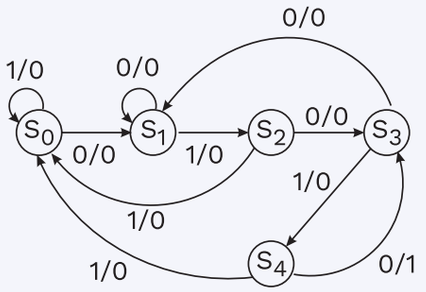


Fig. 6.145

(A) The sequence 01110 is detected.
(B) The sequence 01001 is detected.
(C) The sequence 01011 is detected.
(D) The sequence 01010 is detected.

Solution: (D)

Example 31:

Find the reduced state table and reduced state diagram.

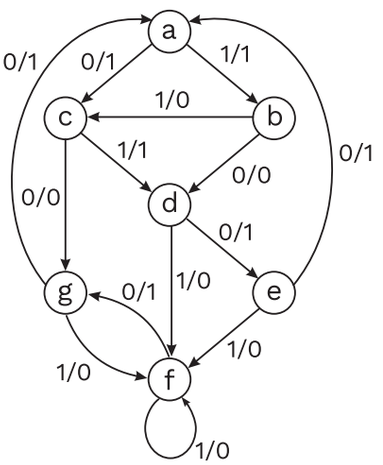


Fig. 6.146

Solution:

State table:

Present State	Next State, Output	
	X = 0	X = 1
a	c, 1	b, 1
b	d, 0	c, 0
c	g, 0	d, 1
d	e, 1	f, 0
e	a, 1	f, 0
f	g, 1	f, 0
g	a, 1	f, 0

Remove State

Table 6.69

- State-g and state-e are the same, so we can remove state-g and where ever g is present, we can replace with state-e.
- State-d and state-f are same, so we can remove state-f and where ever f is present, we can replace it with state-d.

Reduced state table:

Present State	Next State, Output	
	X = 0	X = 1
a	c, 1	b, 1
b	d, 0	c, 0
c	e, 0	d, 1
d	e, 1	d, 0
e	a, 1	d, 0

Table 6.70

Reduced state diagram:

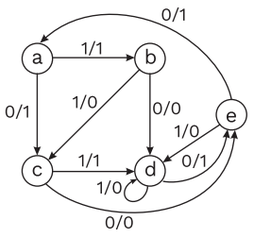


Fig. 6.147

Sequence Detector

A sequence detector is sequential machine which produces an output '1' every time the desired sequence is detected and an output '0' at all other times.

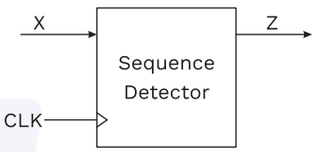


Fig. 6.148

1. There are two types of sequence detectors-
   a) Overlapping sequence detector
   b) Non over lapping sequence detector

Example 32:

Find the output of the non-overlapping sequence detector to detect 1101 for the input sequence X = 1101101101.

Solution:

Given, input sequence,

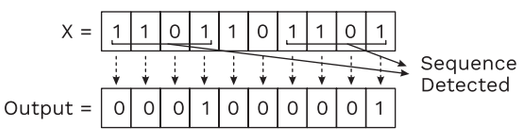


Fig. 6.149

Example 33:

Find the output of the overlapping sequence detector to detect 1101 for the input sequence X = 1101101101.

Solution:

Given input sequence,

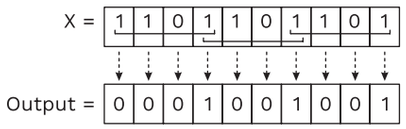


Fig. 6.150

Example 34:

Find the output of the non-overlapping sequence detector to detect 11011 for the input sequence X = 1101101101.

Solution:

Given input sequence,

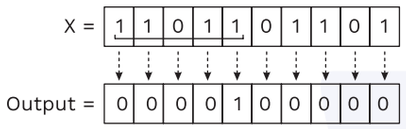


Fig. 6.151

Example 35:

Find the output of the overlapping sequence detector to detect 11011 for the input sequence X = 1101101101.

Solution:

Given input sequence,

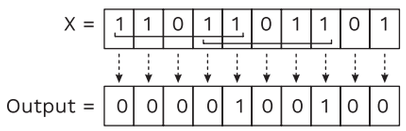
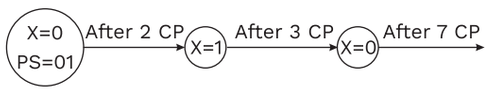


Fig. 6.152

Example 36:

Consider the state diagram find,

1. If X = 0, present state = 11, then next state after 10 clock pulses.
2. If X = 1, present state = 01, then next state after 4 clock pulses.

3. 

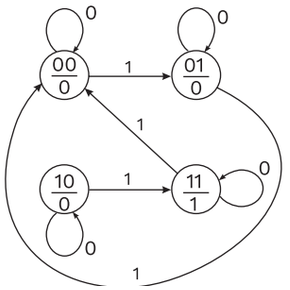
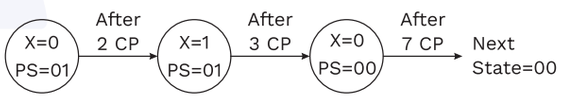


Fig. 6.153

Solution:

1. If input X = 0, Present state = 11
   Next state after 1 clock pulse = 11
   Next state after 2 cock pulse = 11
   ‘‘ ‘‘ ‘‘ ‘‘
   ‘‘ ‘‘ ‘‘ ‘‘
   ‘‘ ‘‘ ‘‘ ‘‘
   ‘‘ ‘‘ ‘‘ ‘‘
   Next state after 10 clock pulse = 11
   So, for input X = 0, after 10 clock pulse next state = present state = 11
2. If input X = 1, present state = 01
   Next state after 1 clock pulse = 00
   Next state after 2 clock pulse = 01
   Next state after 3 clock pulse = 00
   Next state after 4 clock pulse = 01
   So, for input X=1, after 4 clock pulse next state = 01
3. 

State diagram to detect the sequence:

We can develop the state diagram to detect the given sequence by using two models.

1. Mealy model
2. Moore model

Mealy model: To detect n-bit sequence by using Mealy model n number of states required.

Moore model: To detect n-bit sequence by using Moore model (n+1) number of states are required.

Example 37:

Develop the non-overlapping sequence detector to detect 1101 by using Mealy model.

Solution:

Detection of 1101, there should be 4-states. Lets state are a, b, c and d.

For 1101 detection,

- At state-a

  → If first bit is '0' → create self loop
  → Input = 0 & Output = 0

  → If first bit is '1' → move to state-b
  → Input = 1 & Output = 0

- At state-b

  → If second bit is '0' → move back to state-a
  → Input = 0 & Output = 0

  → If second bit is '1' → move to state-c
  → Input = 1 & Output = 0

- At state-c

  → If third bit is '0' → move to state-d
  → Input = 0 & Output = 0

  → If third bit is '1' → create self loop
  → Input = 1 & Output = 0

- At state-d

  → If fourth bit is '0' → reset and move to state-a
  → Input = 0 & Output = 0

  → If fourth bit is '1' → final bit detected and move to state-a → Input = 1 & Output = 1

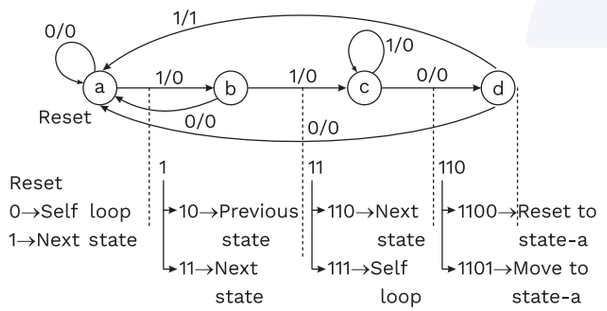


Fig. 6.154

Example 38:

Develop the non-overlapping sequence detector to detect 1010 by using Mealy model.

Solution:

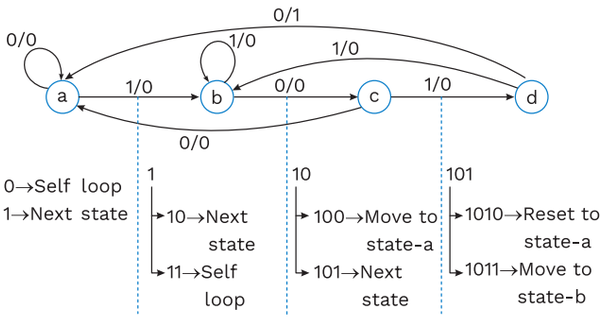


Fig. 6.155

Example 39:

Develop the over lapping sequence detector to detect 1101 by using Mealy model.

Solution:

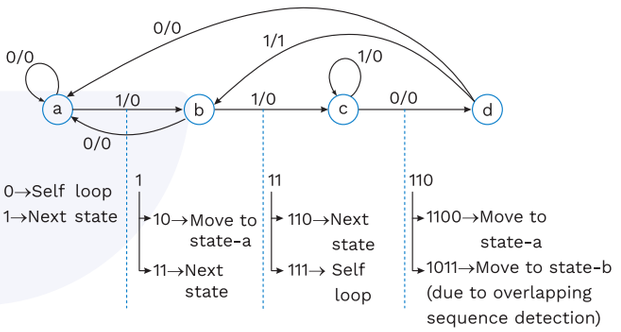


Fig. 6.156

- Overlapping sequence detector stores previous bit.

Example 40:

Develop the overlapping sequence detector to detect 1010 by using Mealy model.

Solution:

In overlapping sequence detector, it stores the previous bit.

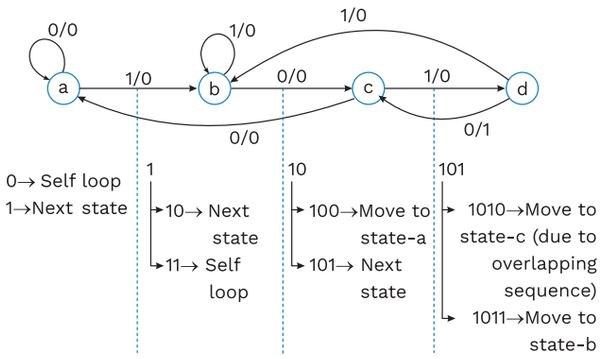


Fig. 6.157

Example 41:

Develop the non-overlapping sequence detector to detect 1101 by using the Moore model.

Solution:

In the Moore model, to detect n-bit sequence, we required (n+1) states,

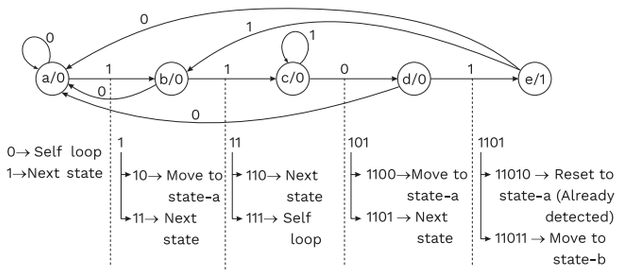


Fig. 6.158

Example 42:

Develop the non-overlapping sequence detector and state table to detect 1010 by using the Moore model.

Solution:

We require (n+1) states,

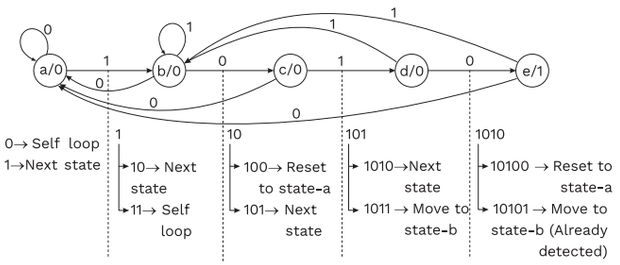


Fig. 6.159

Example 43:

Develop the overlapping sequence detector to detect 1101 by using the Moore model.

Solution:

We require 5 states, in overlapping it stores a previous bit.

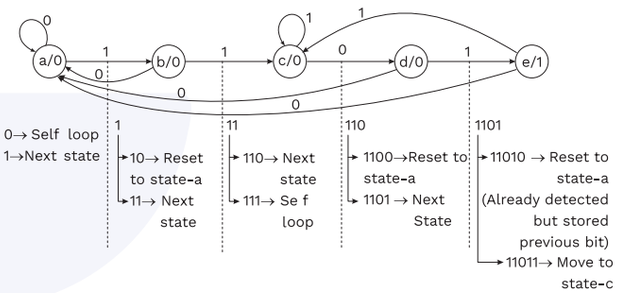
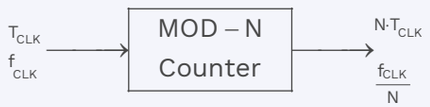


Fig. 6.160

Chapter Summary

- In sequential circuits memory unit is required to store the past history of the input.
- Latch can store 1-bit of data and work as a memory element.
- Clear control is used to reset the content of the flip-flop output.
- Present control is used to set the content of flip-flop output.
- Race-around condition always arises in 'Asynchronous circuits'.
- D-flip-flop is used to provide a delay in digital circuits.
- There are two types of triggering:
  - Level triggering
  - Edge triggering
- Condition to eliminate race around the condition.
  $(T_{CLK})_{ON} < (t_{pd})_{FF} < T$
- The state which is counted by the counter is called valid states.
- The states which are skipped by the counter are called invalid states.
- The minimum number of clock cycles needed to get the counting pattern to repeat itself is called as modulus of a counter.
- 

- In the Asynchronous counter, flip-flops are always operated in toggle mode.
- Total delay for n-bit ripple counter = ntpd.
- The strobe signal is used to eliminate transition states.
- In binary counter, there will be no invalid states.
- In non-binary counter, there will be one on more invalid states.
- For the Asynchronous counter,
  TCLK ≥ ntpd

  $$f_{CLK} \leq \frac{1}{nt_{pd}}$$
- A ring counter is asynchronous counter which works as a shift register.
- In "Moore state machine" present output depends only on the present state.
- In "Mealy state machine" present output depends upon the present state and present external inputs.

Practice Questions

1. AB flip-flop whose characteristic table in given below is to be implemented using SR flip-flop, then find the value of S and R.

A	B	Q^+
0	0	1
0	1	0
1	0	0
1	1	1

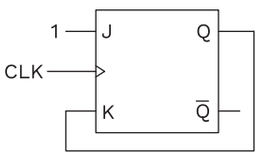
(A) $S = \overline{A \oplus B}$ and R = AB

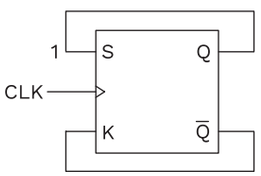
(B) S = A Å B and R = A ⊙ B

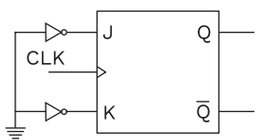
(C) $S = \bar{A}B$ and R = AB

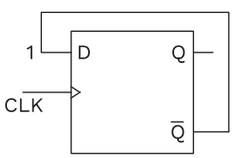
(D) $S = \overline{A \oplus B}$ and $R = \overline{A \odot B}$

2. Which of the following flip-flops define toggle mode:

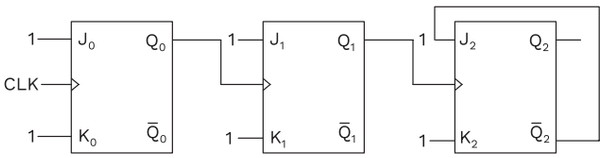
(A) 

(B) 

(C) 

(D) 

3. Find the frequency of output Q_1 and Q_2 shown in the figure; if the clock frequency is 100 kHz.



(A) $f_{Q1} = 100kHz; f_{Q2} = 100kHz$

(B) $f_{Q1} = 200kHz; f_{Q2} = 400kHz$

(C) $f_{Q1} = 50kHz; f_{Q2} = 250kHz$

(D) $f_{Q1} = 50kHz; f_{Q2} = \frac{100}{3}kHz$

4. What is the MODULUS of the counter for the counting sequence: 00 → 10 → 01 → 11 → 01 → 10 → 00 → 10 → 01 → 11 → 01 → 10

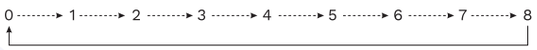
(A) 6

(B) 4

(C) 5

(D) 7

5. Counter counting states are given below; find which counter satisfies these states.



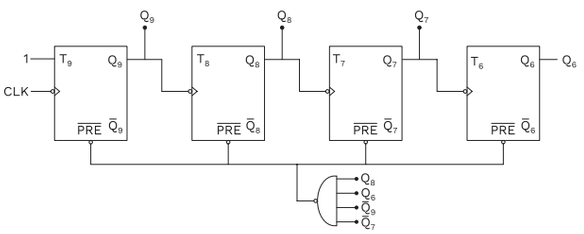
(A) 3-Bit Non-binary Up-counter

(B) 3-Bit Binary Up-counter

(C) 4-Bit Binary Up-counter

(D) 4-Bit Non-binary Up-counter

6. An asynchronous up counter is given below, find the MOD of the counter circuit.



(A) MOD-10

(B) MOD-6

(C) MOD-11

(D) MOD-5

7. A 6-bit ring counter has how many unused state?

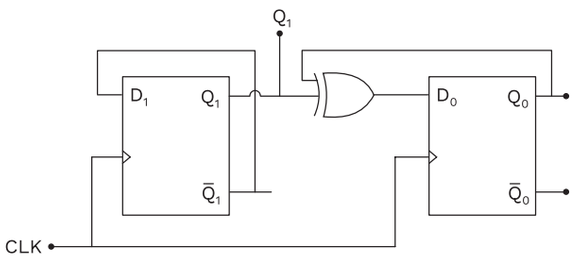
(A) 58

(B) 70

(C) 38

(D) 26

8. A synchronous counter is shown below; what will be the counting sequence for the counter circuit.



(A) 00 → 01 → 11 → 10 → 00 → 01 → ....

(B) 00 → 11 → 10 → 01 → 00 → 11 → ....

(C) 00 → 10 → 01 → 11 → 00 → 10 → ....

(D) 00 → 10 → 11 → 101 → 00 → 10 → ....

9. Find the output of the overlapping sequence detector to detect 101 for the input sequence X = 11010100100010101,

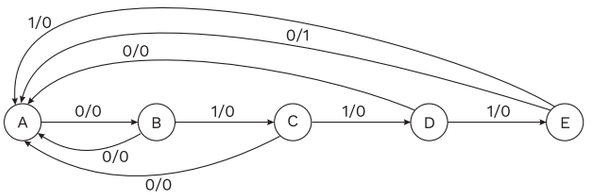
(A) 000101000101000101

(B) 000101000010000101

(C) 010100101000010010

(D) 000100101001001001

10. The state diagram of the sequence detector is shown below; if the output is 1, then:



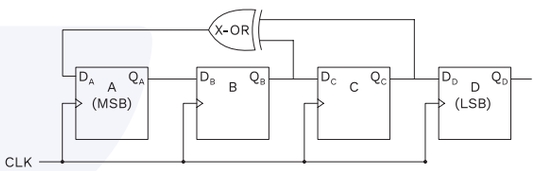
(A) The sequence 01111 is detected.

(B) The sequence 00001 is detected.

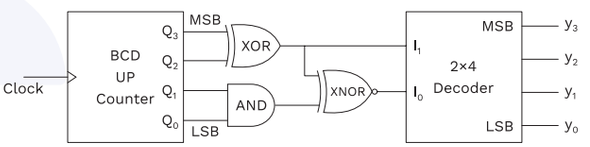
(C) The sequence 01110 is detected.

(D) The sequence 10111 is detected.

11. The counter shown below is initially at 1001. The Modulus number for the given counter is _____.



12. Consider the logic diagram shown in the figure below,



the duty cycle at y3 is _____ %.

Practice Solutions

1. Solution: (D) **2. Solution: (A), (C) and (D)** **3. Solution: (C)** **4. Solution: (A)**

5. Solution: (D) **6. Solution: (D)** **7. Solution: (A)** **8. Solution: (C)**

9. Solution: (B) **10. Solution: (C)** **11. Solution: 7** **12. Solution: 10**